Formula and Quantitative Estimation Handbook

From first principles to expert practice

Built from the manuscript sources · 0 chapter(s) in this edition

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Appendix C — Formula and Quantitative Estimation Handbook

> Generated by `build/build.py` from every `formula` callout in the manuscript, grouped by chapter. Worked examples remain in the chapters; follow the section reference.

Chapter 1 — What Is System Design?

From §1.3.2

FORMULA

Average and peak request rate

$$\text{RPS}_{\text{avg}} = \frac{\text{DAU} \times r}{86,400} \qquad \text{RPS}_{\text{peak}} = k \times \text{RPS}_{\text{avg}}$$

Symbol	Meaning	Units
RPS_{avg}	Average request rate over a day	requests/second
DAU	Daily active users	users
r	Requests (of the type being estimated) per user per day	requests/user/day
$86,400$	Seconds in a day	seconds/day
k	Peak factor: ratio of the busiest period's rate to the daily average (dimensionless; 2–5 is a common assumption for consumer apps)	—
RPS_{peak}	Request rate during the busiest period	requests/second

From §1.7.2

FORMULA

Availability and downtime budget

$$A = \frac{\text{MTBF}}{\text{MTBF}+\text{MTTR}} \qquad D = (1 - A) \times T$$

Symbol	Meaning	Units
A	Availability: fraction of time the service is up (dimensionless, 0–1)	—
MTBF	Mean time between failures	hours
MTTR	Mean time to repair (restore service)	hours
D	Allowed downtime in the period	minutes (or hours)
T	Length of the period	minutes (or hours)

From §1.7.2

FORMULA

Availability of components in series (independent failures)

$$A_{\text{series}} = \prod_{i=1}^{n} A_i$$

Symbol	Meaning	Units
A_{series}	Availability of the whole request path	—
A_i	Availability of component i	—
n	Number of components the request depends on	count

From §1.8.1

FORMULA

Weighted decision score

$$S_j = \sum_{i=1}^{m} w_i \, s_{ij}, \qquad \sum_{i=1}^{m} w_i = 1$$

Symbol	Meaning	Units
S_j	Total score of option j	points (same scale as s_{ij})
w_i	Weight of criterion i (importance; sums to 1)	—
s_{ij}	Score of option j on criterion i (here 1 = poor, 5 = excellent)	points
m	Number of criteria	count

Chapter 2 — How to Think Like a System Designer

From §2.5.1

FORMULA

Allowed downtime for an availability target

$$T_{down} = (1 - A) \times T_{period}$$

Symbol	Meaning	Units
T_{down}	Maximum downtime allowed in the period	minutes
A	Availability target, as a fraction	dimensionless (e.g. 0.9995)
T_{period}	Length of the measurement period	minutes

From §2.6.1

FORMULA

Service demand, the utilization law and the bottleneck bound

$$D_i = V_i \times S_i \qquad U_i = X \times D_i \qquad X_{max} = \frac{c_i}{D_{max}}$$

Symbol	Meaning	Units
D_i	Service demand: total busy time one request places on resource i	seconds per request
V_i	Visits: how many times one request uses resource i	visits per request
S_i	Service time per visit to resource i	seconds per visit
U_i	Utilization of resource i, per unit of capacity	fraction (0–1)
X	System throughput	requests per second
c_i	Number of parallel units of resource i (e.g. cores)	units
X_{max}	Maximum throughput, set by the resource with the largest demand per unit	requests per second

The utilization law for a resource with c_i parallel units is $U_i = X \times D_i / c_i$. The bottleneck is the resource with the largest D_i / c_i.

From §2.6.1

FORMULA

Little’s Law (concurrency)

$$L = \lambda \times W$$

Symbol	Meaning	Units
L	Average number of requests in the system (or holding a resource) at once	requests
λ	Average arrival rate	requests per second
W	Average time each request spends in the system (or holds the resource)	seconds

Little’s Law holds for any stable system, regardless of the arrival pattern [Little 1961]. Full treatment in §4.11.

From §2.8.1

FORMULA

Weighted decision score

$$S_j = \sum_{i=1}^{n} w_i \times s_{ij}, \qquad \sum_{i=1}^{n} w_i = 1$$

Symbol	Meaning	Units
S_j	Total weighted score of option j	score points (same scale as s_{ij})
w_i	Weight of criterion i	fraction (0–1)
s_{ij}	Score of option j on criterion i	score points, e.g. 1–5
n	Number of criteria	count

From §2.9.1

FORMULA

Availability of components in series

$$A_{series} = \prod_{i=1}^{n} A_i$$

Symbol	Meaning	Units
A_{series}	Availability of a request path that needs *all* n components	fraction (0–1)
A_i	Availability of component i	fraction (0–1)
n	Number of components on the synchronous path	count

This assumes independent failures. Parallel (redundant) components and correlated failures are treated in §4.14.

From §2.11.1

FORMULA

Average and peak request rate

$$R_{avg} = \frac{U \times a}{86,400} \qquad R_{peak} = \frac{U \times a \times h}{3,600} \times b$$

Symbol	Meaning	Units
R_{avg}	Average request rate over a day	requests/s
R_{peak}	Design peak request rate	requests/s
U	Daily active users	users
a	Actions of this type per user per day	requests/user/day
h	Share of daily volume in the busiest hour	fraction
b	Burst factor above the busiest-hour average	dimensionless
86,400 / 3,600	Seconds per day / per hour	s

From §2.11.1

FORMULA

Storage and node count

$$S = n_{day} \times s \times T \times k \qquad N = \lceil \frac{R_{peak}}{c \times u} \rceil + N_{spare}$$

Symbol	Meaning	Units
S	Total storage needed	bytes
n_{day}	Records created per day	records/day
s	Size per record, including indexes and overhead	bytes
T	Retention period	days
k	Number of stored copies (replication)	dimensionless
N	Nodes needed	nodes
c	Throughput one node sustains	requests/s
u	Target utilization, leaving head-room for queueing (§4.10)	fraction
N_{spare}	Extra nodes so a failure or deploy does not overload the rest	nodes

Chapter 3 — The System Design Learning Method

From §3.3.2

FORMULA

Fraction of keys remapped when growing from N to $N+1$ nodes

$$f_{\text{mod}} = 1 - \frac{1}{N+1} = \frac{N}{N+1} \qquad f_{\text{ring}} \approx \frac{1}{N+1}$$

Symbol	Meaning	Units
N	Number of nodes before the change	nodes
f_{mod}	Expected fraction of keys whose owner changes under `hash mod N` (uniform hashing)	dimensionless (0–1)
f_{ring}	Expected fraction of keys whose owner changes under consistent hashing	dimensionless (0–1)

From §3.9.1

FORMULA

Gap priority

$$P_i = (T_i - C_i) \times W_i$$

Symbol	Meaning	Units
P_i	Priority score for part i	points (0–12)
T_i	Target proficiency for part i	0–4 scale
C_i	Current self-rated proficiency (use 0 if $C_i > T_i$ would make P_i negative)	0–4 scale
W_i	Relevance weight: 1 = nice to have, 2 = useful, 3 = essential for role	dimensionless (1–3)

Chapter 4 — Quantitative Foundations for System Design

From §4.3.1

FORMULA

Laws of exponents

$$b^m \times b^n = b^{m+n} \qquad \frac{b^m}{b^n} = b^{m-n} \qquad (b^m)^n = b^{mn}$$

Symbol	Meaning	Units
b	Base (2 for binary quantities, 10 for decimal)	dimensionless
m, n	Exponents	dimensionless

From §4.4.2

FORMULA

Logarithm rules and change of base

$$\log_b (xy) = \log_b x + \log_b y \qquad \log_b (x^k) = k \log_b x \qquad \log_b x = \frac{\ln x}{\ln b} = \frac{\log_{10} x}{\log_{10} b}$$

Symbol	Meaning	Units
b	Base of the logarithm ($b > 0$, $b \neq 1$)	dimensionless
x, y	Positive numbers	dimensionless
k	Any real exponent	dimensionless
$\ln$	Natural logarithm, base $e \approx 2.71828$	—

From §4.4.3

FORMULA

Height of a B-tree

$$h = \lceil \log_f n \rceil = \lceil \frac{\log n}{\log f} \rceil$$

Symbol	Meaning	Units
h	Number of levels from root to leaf (page reads per lookup if nothing is cached)	levels
n	Number of keys indexed	keys
f	Fan-out: child pointers per internal node	children/node
$\lceil \cdot \rceil$	Ceiling: round up to the next integer	—

From §4.5.1

FORMULA

Converting link speed to byte throughput

$$R_{\text{B/s}} = \frac{R_{\text{bps}}}{8} \qquad t_{\text{transfer}} = \frac{8 \times S}{R_{\text{bps}}}$$

Symbol	Meaning	Units
R_{bps}	Link rate	bits/second
$R_{\text{B/s}}$	The same rate expressed in bytes	bytes/second
S	Amount of data to transfer	bytes
t_{transfer}	Ideal transfer time, ignoring protocol overhead and congestion	seconds

From §4.5.3

FORMULA

Average and peak rate from a daily volume

$$\bar{\lambda} = \frac{V_{\text{day}}}{86,400} \qquad \lambda_{\text{peak}} = k \times \bar{\lambda}$$

Symbol	Meaning	Units
V_{day}	Events (requests, writes, messages) per day	events/day
$\bar{\lambda}$	Average rate	events/second
k	Peak-to-average ratio, measured or assumed (often 2–10 for consumer traffic)	dimensionless
λ_{peak}	Rate the system must be sized for	events/second

From §4.6.1

FORMULA

Probability that at least one of N independent components fails

$$P(\text{at least one fails}) = 1 - (1 - p)^N$$

Symbol	Meaning	Units
p	Probability that a single component fails during the period of interest	dimensionless (0–1)
N	Number of independent components	count
$(1 - p)^N$	Probability that *none* fails (complement, multiplied N times by independence)	dimensionless

From §4.6.2

FORMULA

Union (inclusion-exclusion) and intersection

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A \cap B) = P(A) \times P(B \mid A)$$

Symbol	Meaning	Units
$P(A \cup B)$	Probability that A or B (or both) happens	dimensionless
$P(A \cap B)$	Probability that both happen	dimensionless
$P(B \mid A)$	Conditional probability of B given that A happened; equals $P(B)$ when independent	dimensionless

From §4.6.3

FORMULA

Expected value and linearity of expectation

$$E[X] = \sum_{i} x_i \, P(X = x_i) \qquad E[X_1 + X_2 + \cdots + X_N] = E[X_1] + \cdots + E[X_N]$$

Symbol	Meaning	Units
X	A random quantity (failures, latency, cost)	units of the quantity
x_i	A possible value of X	units of the quantity
$P(X = x_i)$	Probability of that value	dimensionless

From §4.6.4

FORMULA

Binomial probability of exactly k failures

$$P(K = k) = \binom{N}{k} p^k (1-p)^{N-k}, \qquad \binom{N}{k} = \frac{N!}{k!\,(N-k)!}, \qquad E[K] = Np$$

Symbol	Meaning	Units
N	Number of independent components (trials)	count
k	Number of failures we ask about	count
p	Per-component failure probability in the period	dimensionless
$\binom{N}{k}$	Number of ways to choose which k components fail ("N choose k")	count
$E[K]$	Expected number of failures	count

From §4.7.1

FORMULA

Mean, variance and standard deviation

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \sigma^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2 \qquad \sigma = \sqrt{\sigma^2}$$

Symbol	Meaning	Units
x_i	The i-th measurement	e.g. ms
n	Number of measurements	count
$\bar{x}$	Mean	same as x (ms)
σ^2	Variance: average squared distance from the mean	ms²
σ	Standard deviation: typical distance from the mean	ms

From §4.7.2

FORMULA

Tail formulas for three skewed distributions

Exponential: $P(T > t) = e^{-\lambda t}$, $E[T] = \frac{1}{\lambda}$ Log-normal: median $= e^{\mu}$, mean $= e^{\mu + s^2/2}$ Pareto: $P(X >$

Symbol	Meaning	Units
λ	Event rate of the exponential distribution	events/second
t	Elapsed time	seconds
μ, s	Mean and standard deviation of $\ln X$ for a log-normal X	ln of the unit of X
x_m	Pareto minimum (scale) value	unit of X (e.g. bytes)
α	Pareto tail index: smaller means heavier tail	dimensionless

From §4.8.2

FORMULA

Nearest-rank percentile

$$r = \lceil \frac{q}{100} \times n \rceil, \qquad \mathrm{p}q = x_{(r)}$$

Symbol	Meaning	Units
q	Desired percentile, $0 < q \le 100$	percent
n	Number of samples	count
r	1-based rank in the sorted list	position
$x_{(r)}$	The r-th smallest sample	e.g. ms

From §4.8.4

FORMULA

Probability that a fan-out request is fast

$$P(\text{all } N \text{ fast}) = p^N \qquad P(\text{at least one slow}) = 1 - p^N \qquad p_{\text{required}} = P_{\text{target}}^{1/N}$$

Symbol	Meaning	Units
p	Probability that one leaf responds within the threshold (e.g. 0.99 at the leaf's p99)	dimensionless
N	Number of parallel sub-requests the root must wait for	count
P_{target}	Desired probability that the whole request is fast	dimensionless
p_{required}	Per-leaf probability needed to meet P_{target}	dimensionless

From §4.9.2

FORMULA

Dot product and cosine similarity

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^{d} a_i b_i \qquad \cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \, \|\mathbf{b}\|}$$

Symbol	Meaning	Units
$\mathbf{a}, \mathbf{b}$	Two vectors of the same dimension	(embedding units; dimensionless)
d	Dimension	count
a_i, b_i	i-th components	dimensionless
$\|\mathbf{a}\|$	Length of $\mathbf{a}$	dimensionless
θ	Angle between the vectors	radians or degrees
$\cos\theta$	Cosine similarity: 1 = same direction, 0 = unrelated (perpendicular), −1 = opposite	dimensionless

From §4.9.3

FORMULA

Cost of matrix multiplication

$$\text{FLOPs} = 2mnk \qquad \text{arithmetic intensity} = \frac{\text{FLOPs}}{\text{bytes moved}}$$

Symbol	Meaning	Units
m, k	Rows and columns of A	count
k, n	Rows and columns of B (inner dimension k must match)	count
FLOPs	Floating-point operations: one multiply and one add per term	operations
bytes moved	Data read from and written to memory	bytes
arithmetic intensity	FLOPs per byte; low values mean memory-bandwidth-bound, high values compute-bound	FLOP/byte

From §4.10.2

FORMULA

M/M/1 results

$$\rho = \frac{\lambda}{\mu} \qquad W = \frac{1}{\mu - \lambda} = \frac{S}{1 - \rho} \qquad W_q = \frac{\rho}{\mu - \lambda} \qquad L = \frac{\rho}{1 - \rho} \qquad L_q = \frac{\rho^2}{1 - \rho}$$

Symbol	Meaning	Units
λ	Average arrival rate	jobs/second
μ	Average service rate of the server	jobs/second
$S = 1/\mu$	Average service time	seconds
ρ	Utilization: fraction of time the server is busy (must be < 1)	dimensionless
W	Average time in system (waiting + service), i.e. response time	seconds
W_q	Average time waiting in the queue before service starts	seconds
L	Average number of jobs in the system	jobs
L_q	Average number of jobs waiting in the queue	jobs

From §4.10.4

FORMULA

M/M/c (Erlang C)

$$a = \frac{\lambda}{\mu}, \quad \rho = \frac{a}{c}, \quad P_{\text{wait}} = \frac{\frac{a^c}{c!\,(1-\rho)}}{\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!\,(1-\rho)}}, \quad W = \frac{P_{\text{wait}}}{c\mu - \lambda} + \frac{1}{\mu}$$

Symbol	Meaning	Units
c	Number of servers sharing one queue	count
μ	Service rate of *each* server	jobs/second
a	Offered load: average number of busy servers	servers (Erlangs)
ρ	Per-server utilization (< 1)	dimensionless
P_{wait}	Probability an arriving job finds all servers busy	dimensionless
W	Mean response time	seconds

From §4.11.1

FORMULA

Little’s Law

$$L = \lambda \times W$$

Symbol	Meaning	Units
L	Average number of items in the system (concurrency, occupancy, in-flight requests)	items
λ	Average arrival rate (equals throughput in a stable system)	items/second
W	Average time an item spends inside the system boundary	seconds

From §4.12.1

FORMULA

Amdahl's Law (fixed problem size)

$$S(N) = \frac{1}{(1 - p) + \frac{p}{N}} \qquad \lim_{N \to \infty} S(N) = \frac{1}{1 - p}$$

Symbol	Meaning	Units
p	Fraction of the single-processor run time that can be parallelized	dimensionless (0–1)
$1 - p$	Serial fraction	dimensionless
N	Number of processors, cores or nodes	count
$S(N)$	Speedup relative to one processor	dimensionless (×)

From §4.12.2

FORMULA

Gustafson's Law (scaled problem size)

$$S_{\text{scaled}}(N) = (1 - p) + pN = N - (1 - p)(N - 1)$$

Symbol	Meaning	Units
p	Fraction of the *parallel run's* time spent in parallel work	dimensionless
$1 - p$	Fraction of the parallel run's time that is serial	dimensionless
N	Number of processors	count
S_{scaled}	How much more work is completed in the same time than one processor could do	dimensionless (×)

From §4.13.1

FORMULA

Universal Scalability Law

$$C(N) = \frac{N}{1 + \alpha(N - 1) + \beta N(N - 1)} \qquad N^* = \frac{\sqrt{1 - \alpha}}{\beta}$$

Symbol	Meaning	Units
N	Number of nodes, cores, threads or concurrent users	count
$C(N)$	Relative capacity: throughput at N divided by throughput at 1	dimensionless (×)
α	Contention coefficient: serialization on shared resources (queues, locks)	dimensionless (0–1)
β	Coherency coefficient: pairwise coordination cost (cache coherence, consensus, data exchange)	dimensionless (≥ 0)
N^*	Concurrency at which throughput peaks	count

From §4.14.1

FORMULA

Downtime budget

$$D = (1 - A) \times T$$

Symbol	Meaning	Units
A	Availability target	dimensionless
T	Length of the period	minutes (or any time unit)
D	Allowed downtime in the period (the error budget, in time)	same unit as T

From §4.14.2

FORMULA

Series, parallel and k-of-n availability (independent failures)

$$A_{\text{series}} = \prod_{i=1}^{n} A_i \qquad A_{\text{parallel}} = 1 - \prod_{i=1}^{n} (1 - A_i) \qquad A_{k\text{-of-}n} = \sum_{j=k}^{n} \binom{n}{j} A^j (1 - A)^{n-j}$$

Symbol	Meaning	Units
A_i	Availability of component i	dimensionless
$\prod$	Product over all components	—
n	Number of components	count
k	Minimum number of identical components that must be up	count
A	Availability of each identical component (k-of-n case)	dimensionless

From §4.15.1

FORMULA

Reliability and availability from MTBF and MTTR

$$\lambda = \frac{1}{\text{MTBF}} \qquad R(t) = e^{-\lambda t} \qquad A = \frac{\text{MTBF}}{\text{MTBF+MTTR}} \qquad \text{AFR} = 1 - e^{-8760/\text{MTTF}}$$

Symbol	Meaning	Units
MTBF	Mean operating time between failures	hours
MTTR	Mean time to repair (detect + diagnose + fix + restore)	hours
MTTF	Mean time to failure of a non-repairable part	hours
λ	Constant failure rate	failures/hour
t	Mission time: the period we need it to survive	hours
$R(t)$	Probability of no failure during t	dimensionless
A	Steady-state availability	dimensionless
AFR	Annualized failure rate: probability a part fails within a year (8,760 h)	dimensionless (often %)

From §4.16.1

FORMULA

Approximate annual probability of losing an n-way replicated item

$$P_{\text{loss/yr}} \approx n!\ \lambda^{n}\ T^{n-1} \times H \qquad \text{valid when } \lambda T \ll 1$$

Symbol	Meaning	Units
n	Number of replicas	count
λ	Per-replica failure rate (AFR / 8,760 for small AFR)	failures/hour
T	Mean time to detect and re-replicate a lost copy	hours
H	Hours per year (8,760)	hours/year
$P_{\text{loss/yr}}$	Annual probability that all n replicas are lost	dimensionless

From §4.18.4

FORMULA

Standard error of a sample mean

$$\text{SE} = \frac{\sigma}{\sqrt{n}}$$

Symbol	Meaning	Units
σ	Standard deviation of the quantity being measured	units of the quantity
n	Number of independent samples	count
SE	Typical error of the estimated mean	units of the quantity

From §4.18.5

FORMULA

Compute-bound token throughput

$$X_{\text{tokens}} = \frac{\text{MFU} \times F_{\text{peak}}}{2P}$$

Symbol	Meaning	Units
F_{peak}	Peak accelerator throughput at the precision used	FLOP/s
MFU	Model FLOPs utilization	dimensionless (0–1)
P	Model parameters	count
$2P$	Forward-pass FLOPs per token (approximation)	FLOP/token
X_{tokens}	Tokens processed per second per accelerator	tokens/s

From §4.18.6

FORMULA

Batching model

$$T(B) = t_0 + B\,t_1 \qquad X(B) = \frac{B}{t_0 + B\,t_1} \qquad W_{\text{fill}} \approx \frac{B-1}{2\lambda}$$

Symbol	Meaning	Units
B	Batch size	items
t_0	Fixed cost per batch	seconds
t_1	Marginal cost per item	seconds/item
$T(B)$	Time to process one batch	seconds
$X(B)$	Throughput	items/second
λ	Arrival rate of items	items/second
W_{fill}	Average time an item waits for its batch to fill	seconds

Chapter 5 — Computer Architecture

From §5.3.2

FORMULA

CPU execution time (the “iron law”)

$$T_{CPU} = N_{instr} \times CPI \times t_{cycle} = \frac{N_{instr} \times CPI}{f}$$

Symbol	Meaning	Units
T_{CPU}	CPU time to run the program	seconds
N_{instr}	Number of instructions executed	instructions
CPI	Average clock cycles per instruction (its inverse is IPC, instructions per cycle)	cycles/instruction
t_{cycle}	Duration of one clock cycle	seconds/cycle
f	Clock frequency, $1/t_{cycle}$	cycles/second (Hz)

From §5.4.4

FORMULA

Address decomposition in a set-associative cache

$$S = \frac{C}{B \times W}, \qquad \text{offset bits} = \log_2 B, \qquad \text{index bits} = \log_2 S$$

$$\text{set index} = \left\lfloor \frac{A}{B} \right\rfloor \bmod S, \qquad \text{tag} = \left\lfloor \frac{A}{B \times S} \right\rfloor$$

Symbol	Meaning	Units
C	Cache capacity	bytes
B	Line (block) size	bytes
W	Associativity (ways per set)	lines
S	Number of sets	sets
A	Memory address	byte address
tag	High-order address bits stored with the line to identify it	—

From §5.4.5

FORMULA

Average memory access time (AMAT), applied recursively to each level

$$AMAT = t_{L1} + m_{L1} \times (t_{L2} + m_{L2} \times (t_{L3} + m_{L3} \times t_{mem}))$$

Symbol	Meaning	Units
t_{L1}, t_{L2}, t_{L3}	Hit time at each cache level	ns
t_{mem}	DRAM access time	ns
m_{L1}	Fraction of all accesses that miss L1	dimensionless (0–1)
m_{L2}, m_{L3}	**Local** miss rate: fraction of accesses *reaching* that level that miss it	dimensionless (0–1)

From §5.6.1

FORMULA

HDD access time

$$T_{access} = T_{seek} + T_{rot} + T_{transfer}, \qquad T_{rot} = \frac{1}{2} \times \frac{60}{RPM}, \qquad T_{transfer} = \frac{S}{R}$$

$$IOPS_{random} \approx \frac{1}{T_{access}}$$

Symbol	Meaning	Units
T_{seek}	Average time to move the head to the target track	ms
T_{rot}	Average rotational latency: half a revolution	ms
RPM	Spindle speed	revolutions per minute
S	Size of the request	bytes
R	Sustained media transfer rate	bytes/second
$IOPS_{random}$	Random I/O operations per second for one outstanding request	operations/second

From §5.6.3

FORMULA

Write amplification factor and drive lifetime

$$WAF = \frac{\text{bytes written to flash}}{\text{bytes written by the host}}, \qquad \text{Lifetime (days)} \approx \frac{C \times PE}{W_{host} \times WAF}$$

Symbol	Meaning	Units
WAF	Write amplification factor (≥ 1 for a drive without compression)	dimensionless
C	Raw flash capacity	bytes
PE	Rated program/erase cycles per block	cycles
W_{host}	Host write volume per day	bytes/day

From §5.6.5

FORMULA

Little’s Law applied to a storage device

$$IOPS = \frac{QD}{L}, \qquad \text{Throughput} = IOPS \times S$$

Symbol	Meaning	Units
QD	Average number of requests outstanding (queue depth)	requests
L	Average latency per request	seconds
S	Request size	bytes

From §5.8.2

FORMULA

Thread pool sizing heuristic [Goetz et al. 2006, §8.2]

$$N_{threads} = N_{cores} \times U_{target} \times (1 + \frac{W}{C})$$

Symbol	Meaning	Units
$N_{threads}$	Threads in the pool	threads
N_{cores}	Cores available to the pool	cores
U_{target}	Target CPU utilization (0–1)	dimensionless
W	Average time a task spends waiting (I/O, remote calls)	seconds
C	Average CPU time per task	seconds

From §5.9.1

FORMULA

Peak DRAM bandwidth

$$BW_{peak} = R_{transfer} \times W_{bus} \times N_{channels}$$

Symbol	Meaning	Units
$R_{transfer}$	Data transfers per second per channel (the “4800” in DDR5-4800)	transfers/second (MT/s)
W_{bus}	Width of a channel’s data path: 8 bytes (64 bits)	bytes/transfer
$N_{channels}$	Populated memory channels	channels

From §5.12.1

FORMULA

Throughput with batching (fixed plus per-item cost)

$$X(B) = \frac{B}{F + B \times v}, \qquad T_{batch}(B) = F + B \times v, \qquad X_{max} = \frac{1}{v}$$

Symbol	Meaning	Units
B	Items per batch	items
F	Fixed cost per operation (system call, device round trip, interrupt, network RTT)	seconds
v	Variable cost per item	seconds/item
$X(B)$	Throughput at batch size B	items/second
$T_{batch}(B)$	Time to complete one batch	seconds

Chapter 6 — Operating Systems

From §6.4

FORMULA

Context-switch overhead fraction

$$f = \frac{C}{Q + C}$$

Symbol	Meaning	Units
f	Fraction of CPU time lost to switching	dimensionless (0–1)
C	Effective cost of one switch, direct + indirect	µs
Q	Average time a thread runs between switches	µs

From §6.4

FORMULA

Cache refill estimate (indirect cost)

$$C_{\text{refill}} \approx \frac{W}{L} \times \frac{t_{\text{miss}}}{M}$$

Symbol	Meaning	Units
C_{refill}	Time spent reloading the working set after a switch	ns
W	Working set that was evicted	bytes
L	Cache line size (64 on x86-64 and most ARM servers)	bytes/line
t_{miss}	Latency of one miss to the next memory level	ns
M	Memory-level parallelism: misses overlapped in flight	dimensionless (≥ 1)

From §6.5.2

FORMULA

Average turnaround and response time

$$\bar{T}_{\text{turn}} = \frac{1}{n} \sum_{i=1}^{n} \left(T^{i}_{\text{completion}} - T^{i}_{\text{arrival}} \right) \qquad \bar{T}_{\text{resp}} = \frac{1}{n} \sum_{i=1}^{n} \left(T^{i}_{\text{first run}} - T^{i}_{\text{arrival}} \right)$$

Symbol	Meaning	Units
n	Number of jobs	jobs
T^{i}_{arrival}	Time job i became ready	ms
$T^{i}_{\text{first run}}$	Time job i first got a CPU	ms
$T^{i}_{\text{completion}}$	Time job i finished	ms

From §6.5.5

FORMULA

Proportional CPU share

$$s_i = \frac{w_i}{\sum_{j \in R} w_j}$$

Symbol	Meaning	Units
s_i	Fraction of one core's time thread i receives when all threads in R compete	dimensionless (0–1)
w_i	Weight of thread i, from its nice value (nice 0 = 1024; each nice step changes weight by about 1.25×)	dimensionless
R	Set of runnable threads competing on that core	—

From §6.6.3

FORMULA

TLB reach

$$R_{\text{TLB}} = E \times P$$

Symbol	Meaning	Units
R_{TLB}	Amount of memory addressable without a TLB miss	bytes
E	Number of TLB entries	entries
P	Page size per entry	bytes

From §6.6.3

FORMULA

Effective memory access time with page faults

$$\mathrm{EAT} = (1 - p) \times t_{\mathrm{mem}} + p \times t_{\mathrm{fault}}$$

Symbol	Meaning	Units
EAT	Average time per memory access	ns
p	Probability that an access causes a major page fault	dimensionless (0–1)
t_{mem}	Time for an access with no fault	ns
t_{fault}	Time to service a major fault, including the disk read	ns

From §6.14.3

FORMULA

Group-commit throughput

$$X = \frac{B}{t_{\mathrm{fsync}}}$$

Symbol	Meaning	Units
X	Maximum durable commits per second on one log	commits/s
B	Commits made durable by one `fsync` (batch size)	commits
t_{fsync}	Latency of one `fsync`	s

From §6.15.6

FORMULA

Memory for concurrent connections

$$M_{\text{total}} = N \times (m_{\text{unit}} + m_{\text{conn}})$$

Symbol	Meaning	Units
M_{total}	Memory needed for concurrency overhead	bytes
N	Concurrent connections (or in-flight requests)	connections
m_{unit}	Resident memory per concurrency unit (thread stack + kernel task, or coroutine)	bytes
m_{conn}	Per-connection buffers (socket buffers, TLS state, request object)	bytes

From §6.15.6

FORMULA

Thread-pool sizing

$$N_{\text{threads}} = \lambda \times W \qquad \text{and, for mixed CPU/I/O work,} \qquad N_{\text{threads}} \approx N_{\text{cpu}} \times U \times (1 + \frac{T_{\text{wait}}}{T_{\text{cpu}}})$$

Symbol	Meaning	Units
N_{threads}	Threads needed to sustain the load	threads
λ	Request arrival rate	requests/s
W	Time a request occupies a thread	s
N_{cpu}	Available cores	cores
U	Target CPU utilization	dimensionless (0–1)
T_{wait}, T_{cpu}	Per-request time spent waiting and computing	ms

Chapter 7 — Networking Fundamentals

From §7.3.3

FORMULA

Protocol efficiency (goodput fraction)

$$\eta = \frac{P}{P + H}$$

Symbol	Meaning	Units
η	Fraction of transmitted bytes that are application payload	dimensionless (0–1)
P	Application payload bytes per frame	bytes
H	Total per-frame overhead from all layers (headers, trailers, and on Ethernet the preamble and inter-frame gap)	bytes

From §7.4.2

FORMULA

Addresses in a CIDR block

$$N_{\text{addr}} = 2^{W - n} \qquad N_{\text{hosts,IPv4}} = 2^{32 - n} - 2$$

Symbol	Meaning	Units
N_{addr}	Total addresses in the block	addresses
W	Address width: 32 for IPv4, 128 for IPv6	bits
n	Prefix length	bits
$N_{\text{hosts,IPv4}}$	Assignable IPv4 hosts on a traditional subnet: subtract the **network address** (all host bits 0) and the **broadcast address** (all host bits 1)	addresses

From §7.7.5

FORMULA

Retransmission timeout (RFC 6298)

On each new RTT sample R (after the first):

$$\text{RTTVAR} \leftarrow (1-\beta)\,\text{RTTVAR} + \beta\,|\text{SRTT} - R|$$

$$\text{SRTT} \leftarrow (1-\alpha)\,\text{SRTT} + \alpha R$$

$$\text{RTO} = \text{SRTT} + \max(G,\ 4\,\text{RTTVAR})$$

Symbol	Meaning	Units
R	Latest measured round-trip time	ms
SRTT	Smoothed (exponentially averaged) RTT	ms
RTTVAR	Smoothed RTT variation	ms
α, β	Smoothing gains, $1/8$ and $1/4$	dimensionless
G	Clock granularity	ms
RTO	Retransmission timeout; RFC 6298 says round up to at least 1 s	ms

From §7.7.7

FORMULA

CUBIC window growth

$$W(t) = C\,(t-K)^3 + W_{\max}, \qquad K = \sqrt[3]{\frac{W_{\max}\,(1-\beta)}{C}}$$

Symbol	Meaning	Units
$W(t)$	Congestion window t seconds after the last loss	segments
$W_{\max}$	Window just before the last loss	segments
β	Multiplicative decrease factor; window becomes $\beta W_{\max}$ after loss (0.7)	dimensionless
C	Scaling constant (0.4)	segments/s³
K	Time to climb back to $W_{\max}$	s

From §7.15.1

FORMULA

One-hop delay

$$d_{\text{hop}} = \underbrace{\frac{D}{v}}_{\text{propagation}} + \underbrace{\frac{L}{R}}_{\text{transmission}} + d_{\text{queue}} + d_{\text{proc}}$$

Symbol	Meaning	Units
D	Physical length of the link	m (or km)
v	Signal speed in the medium: about 2×10^8 m/s in optical fiber (roughly two-thirds of c)	m/s
L	Packet size	bits
R	Link rate (bandwidth)	bits/s
d_{queue}	Queuing delay: bytes waiting ahead ÷ link rate	s
d_{proc}	Processing delay: microseconds in software, sub-microsecond in switch hardware	s

From §7.15.2

FORMULA

Window-limited throughput and the bandwidth-delay product (BDP)

$$\text{Throughput} \le \frac{W}{\text{RTT}} \qquad \text{BDP} = R \times \text{RTT}$$

Symbol	Meaning	Units
W	Effective window, $\min(\text{cwnd}, \text{rwnd})$	bytes
RTT	Round-trip time	s
R	Bottleneck link rate	bits/s (divide by 8 for bytes/s)
BDP	Data that must be in flight to keep the link full	bits or bytes

From §7.15.3

FORMULA

Mathis throughput formula

$$\text{Throughput} \leq \frac{\text{MSS}}{\text{RTT}} \times \frac{C}{\sqrt{p}}$$

Symbol	Meaning	Units
MSS	Maximum segment size	bits (or bytes, for bytes/s)
RTT	Round-trip time	s
p	Packet loss probability	dimensionless
C	Constant, $\sqrt{3/2} \approx 1.22$ for periodic loss with every packet ACKed	dimensionless

From §7.16.2

FORMULA

Connections needed for a one-request-per-connection protocol

$$N = \lambda \times W$$

Symbol	Meaning	Units
N	Average connections busy	connections
λ	Request rate	requests/s
W	Time a request holds a connection (network RTT plus server time)	s

From §7.17.1

FORMULA

Oversubscription ratio at a leaf

$$\text{Oversubscription} = \frac{\text{total server-facing bandwidth}}{\text{total uplink bandwidth}}$$

Symbol	Meaning	Units
Server-facing bandwidth	Sum of the leaf's downlink port speeds	Gbps
Uplink bandwidth	Sum of the leaf's spine-facing port speeds	Gbps

Chapter 8 — Web Application Fundamentals

From §8.5.1

FORMULA

End-to-end latency of a request on a new connection

$$T_{\text{total}} = T_{\text{DNS}} + T_{\text{TCP}} + T_{\text{TLS}} + T_{\text{req}} + T_{\text{server}} + T_{\text{transfer}}$$

with $T_{\text{TCP}} \approx 1 \times RTT$, $T_{\text{TLS 1.3}} \approx 1 \times RTT$, $T_{\text{req}} \approx 1 \times RTT$ and $T_{\text{transfer}} \approx \frac{S}{B}$ for a response that fits in the congestion window.

Symbol	Meaning	Units
T_{DNS}	DNS lookup time (≈ 0 if cached)	ms
RTT	Client-to-load-balancer round-trip time	ms
T_{server}	Time from the load balancer receiving the request to sending the response (app + DB)	ms
S	Response size	bytes
B	Bottleneck bandwidth	bytes/ms

From §8.6.5

FORMULA

Rows touched per page

$$R_{\text{offset}} = o + \ell \qquad R_{\text{keyset}} \approx \ell + \log_b n$$

Symbol	Meaning	Units
o	Offset (rows skipped)	rows
ℓ	Page size (limit)	rows
n	Rows in the table or index	rows
b	B-tree fan-out (entries per index page, typically hundreds)	entries

From §8.9.2

FORMULA

Varint length

$$\text{bytes}(n) = \max\left(1, \left\lceil \frac{\lfloor \log_2 n \rfloor + 1}{7} \right\rceil\right)$$

Symbol	Meaning	Units
n	Non-negative integer being encoded	—
$\lfloor \log_2 n \rfloor + 1$	Number of significant bits in n (for $n \geq 1$)	bits
7	Payload bits per varint byte	bits/byte

Chapter 9 — Real-Time Communication

From §9.3.1

FORMULA

Short-polling request rate

$$R = \frac{N}{T}$$

Symbol	Meaning	Units
R	Request rate arriving at the servers	requests/second
N	Number of concurrently active (polling) clients	clients
T	Polling interval	seconds

From §9.3.1

FORMULA

Fraction of wasted (empty) polls and added latency

If updates for a client arrive as a Poisson process with rate λ, the probability that a given interval contains no update is

$$P_{\text{empty}} = e^{-\lambda T} \approx 1 - \lambda T \quad (\text{when } \lambda T \ll 1)$$

and the average delay between an event occurring and the client seeing it is

$$D_{\text{avg}} = \frac{T}{2} + t_{\text{rtt}}$$

Symbol	Meaning	Units
P_{empty}	Fraction of polls that return nothing	dimensionless (0–1)
λ	Average update rate per client	events/second
T	Polling interval	seconds
D_{avg}	Average notification delay	seconds
t_{rtt}	Network round-trip time for one poll	seconds

From §9.4.1

FORMULA

Long-polling request rate

With Poisson events at rate λ per client and hold timeout T_h, each request lasts $\min(T_h, X)$ where X is the time to the next event. Its mean is

$$E[\text{duration}] = \frac{1 - e^{-\lambda T_h}}{\lambda}, \qquad R = \frac{N}{E[\text{duration}]}, \qquad C_{\text{open}} \approx N$$

Symbol	Meaning	Units
λ	Average update rate per client	events/second
T_h	Server hold timeout	seconds
$E[\text{duration}]$	Mean lifetime of one long-poll request	seconds
N	Number of connected clients	clients
R	Request rate at the servers	requests/second
C_{open}	Concurrently open requests (Little's Law: $R \times E[\text{duration}]$)	requests

From §9.6.3

FORMULA

WebSocket framing overhead per frame

$$H = 2 + L_{\text{ext}} + 4m, \qquad L_{\text{ext}} = \begin{cases} 0 & n \le 125 \\ 2 & 126 \le n \le 65,535 \\ 8 & n \ge 65,536 \end{cases}$$

Symbol	Meaning	Units
H	Header bytes per frame	bytes
n	Payload length	bytes
L_{ext}	Extended-length bytes	bytes
m	1 if masked (client→server), 0 otherwise	dimensionless

From §9.7.2

FORMULA

Per-participant bandwidth by topology

Topology	Upload per participant	Download per participant	Server work
Mesh	$(N-1)\,b$	$(N-1)\,b$	none
SFU	b (or $\approx 1.5b$–$2b$ with simulcast)	$(N-1)\,b$ (less if receivers get lower layers)	forward $N(N-1)$ streams
MCU	b	b	decode N, mix, encode up to N

Symbol	Meaning	Units
N	Participants in the call	participants
b	Bitrate of one participant's stream	Mbit/s

From §9.9.2

FORMULA

Gateway fleet sizing

$$n_{\text{conn,max}} = \min\left(\frac{f \cdot M}{m},\ c_{\text{blast}}\right) \qquad S = Z \cdot \left\lceil \frac{C}{(Z-1)\,n_{\text{conn,max}}} \right\rceil$$

Symbol	Meaning	Units
M	RAM per gateway server	bytes
f	Fraction of RAM budgeted for connections	dimensionless
m	Memory per connection	bytes/connection
c_{blast}	Maximum connections you accept losing in one server failure	connections
$n_{\text{conn,max}}$	Design ceiling of connections per server	connections
C	Peak concurrent connections	connections
Z	Availability zones; the fleet must survive loss of one	zones
S	Total gateway servers	servers

From §9.9.3

FORMULA

Fan-out load

$$D = \mu \times \bar{r}$$

Symbol	Meaning	Units
D	Message deliveries written to sockets	deliveries/second
μ	Messages published	messages/second
$\bar{r}$	Average number of *online* recipients per message	recipients/message

From §9.9.6

FORMULA

Reconnect arrival rate with uniform jitter

$$\lambda_{\text{reconnect}} \approx \frac{n_{\text{lost}}}{W}$$

Symbol	Meaning	Units
$\lambda_{\text{reconnect}}$	Peak reconnect (handshake) rate	connections/second
n_{lost}	Connections dropped by the failure	connections
W	Width of the random delay window for the first reconnect attempt	seconds

Chapter 10 — Security Fundamentals for System Designers

From §10.3.5

FORMULA

Qualitative risk score

$$R = L \times I$$

Symbol	Meaning	Units
R	Risk score used to rank threats	dimensionless (1–9 on a 3-point scale)
L	Likelihood that the threat is exploited within the planning horizon (1 = unlikely, 2 = plausible, 3 = expected)	ordinal score
I	Impact if exploited (1 = minor, 2 = serious, 3 = severe: data breach, financial loss, regulatory)	ordinal score

From §10.4.1

FORMULA

Birthday bound for hash collisions

$$P_{\text{collision}} \approx \frac{k^2}{2^{n+1}}$$

Symbol	Meaning	Units
k	Number of distinct inputs hashed	count
n	Digest length	bits
$P_{\text{collision}}$	Probability that at least two of the k digests are equal (valid while $P \ll 1$)	probability

Consequence: an n-bit hash offers about $n/2$ bits of collision resistance, because P approaches $1/2$ when $k \approx 2^{n/2}$.

From §10.4.2

FORMULA

Random-nonce collision probability for AES-GCM

$$P_{\text{reuse}} \approx \frac{q^2}{2^{97}}$$

Symbol	Meaning	Units
q	Number of messages encrypted under one key with random 96-bit nonces	messages
P_{reuse}	Probability that any two of them share a nonce (birthday bound with $n = 96$)	probability

From §10.4.7

FORMULA

Entropy of a uniformly random secret

$$H = L \cdot \log_2 N$$

Symbol	Meaning	Units
H	Entropy: the number of bits an attacker must effectively guess	bits
L	Length of the secret	symbols
N	Number of equally likely choices per symbol	symbols

Valid only when every symbol is chosen independently and uniformly at random — true for machine-generated tokens, false for human-chosen passwords.

From §10.6.2

FORMULA

Expected offline brute-force time

$$T = \frac{S}{2 \cdot R \cdot G}$$

Symbol	Meaning	Units
T	Expected time to find one password (average: half the space)	seconds
S	Size of the search space: N^L for random passwords, or the size of the attacker's ranked wordlist up to the target	candidates
R	Guesses per second per attacking device for this hash function and cost	guesses/s/device
G	Number of attacking devices (GPUs)	devices

Salting means T is paid **per user**; without salts, one pass covers all users.

From §10.8

FORMULA

Credential misuse window

$$W = \begin{cases} T_{\text{detect}} + T_{\text{rotate}} & \text{long-lived credential} \\ \min\left(T_{\text{TTL}},\ T_{\text{detect}} + T_{\text{revoke}}\right) & \text{short-lived credential} \end{cases}$$

Symbol	Meaning	Units
W	Time during which a leaked credential remains usable	hours
T_{detect}	Time from leak to detection	hours
$T_{\text{rotate}}, T_{\text{revoke}}$	Time to rotate or revoke after detection	hours
T_{TTL}	Lifetime of a short-lived credential	hours

From §10.9.2

FORMULA

KMS request rate under envelope encryption

$$Q_{\text{KMS}} = \frac{\lambda_{\text{obj}}}{B}$$

Symbol	Meaning	Units
Q_{KMS}	KMS requests generated	requests/s
λ_{obj}	Rate of objects (or records) encrypted or decrypted	objects/s
B	Objects protected per KMS call (1 for a DEK per object; larger with DEK caching or per-batch keys)	objects/request

From §10.11

FORMULA

Amplified attack bandwidth

$$B_{\text{victim}} = B_{\text{attacker}} \times A$$

Symbol	Meaning	Units
B_{victim}	Traffic arriving at the victim	bits/s
B_{attacker}	Spoofed request traffic the attacker sends	bits/s
A	Bandwidth amplification factor of the abused protocol (response bytes / request bytes)	dimensionless

Chapter 11 — Relational Databases

From §11.6.3

FORMULA

I/O cost of join algorithms

$$C_{\text{tuple NLJ}} = M + p_R \cdot M \cdot N \qquad C_{\text{block NLJ}} = M + \lceil \frac{M}{B-2} \rceil N$$

$$C_{\text{index NLJ}} = M + p_R \cdot M \cdot c_{\text{idx}} \qquad C_{\text{Grace hash}} \approx 3(M+N) \qquad C_{\text{sort-merge}} \approx 3(M+N) \text{ to } 5(M+N)$$

Symbol	Meaning	Units
M, N	Pages in the outer relation R and inner relation S	pages
p_R	Rows per page of R	rows/page
B	Buffer pages available to the join (one for output, one for the inner scan)	pages
c_{idx}	Page reads to find matching inner rows through an index (descent plus row fetch)	pages/row
C	Total page reads and writes	page I/Os

From §11.7.5

FORMULA

Break-even selectivity for a secondary-index scan

$$s^* = \frac{P \cdot c_{\text{seq}}}{n \cdot c_{\text{rand}}}$$

Symbol	Meaning	Units
s^*	Fraction of rows matched above which a full scan is cheaper	dimensionless
P	Pages in the table	pages
n	Rows in the table	rows
c_{seq}	Cost of a sequential page read	cost units/page
c_{rand}	Cost of a random page read	cost units/page

From §11.8.2

FORMULA

Height of a B+ tree with L entries per leaf

$$h = 1 + \lceil \log_f \lceil \frac{n}{L} \rceil \rceil, \qquad f = \lfloor \frac{u \cdot P}{e_{\text{int}}} \rfloor, \qquad L = \lfloor \frac{u \cdot P}{e_{\text{leaf}}} \rfloor$$

Symbol	Meaning	Units
h	Levels from root to leaf (page reads per lookup with a cold cache)	levels
n	Entries (rows) indexed	rows
P	Page size	bytes
u	Average fraction of each page holding entries (fill factor minus headers)	dimensionless
e_{int}	Bytes per internal entry (separator key + child pointer + overhead)	bytes
e_{leaf}	Bytes per leaf entry (full row for a clustered index; key + pointer for a secondary index)	bytes
f	Fan-out of internal pages	children/page
L	Entries per leaf page	entries/page

From §11.10.1

FORMULA

Size of the join-order search space

$$N_{\text{left-deep}} = n! \qquad N_{\text{bushy}} = \frac{(2(n-1))!}{(n-1)!} \qquad W_{\text{DP}} \approx n \cdot 2^{n-1}$$

Symbol	Meaning	Units
n	Number of tables joined	tables
$N_{\text{left-deep}}$	Join orders where every join's inner input is a base table	plans
N_{bushy}	All binary join trees, including joins of two intermediate results	plans
W_{DP}	Approximate subproblems examined by Selinger-style dynamic programming over left-deep plans	subproblems

From §11.10.2

FORMULA

Cost of a sequential scan with one filter (PostgreSQL)

$$C_{\text{seq}} = P \cdot c_{\text{seq}} + n \cdot c_{\text{tuple}} + n \cdot k \cdot c_{\text{op}}$$

Symbol	Meaning	Units
P	Heap pages (`pg_class.relpages`)	pages
n	Rows (`pg_class.reltuples`)	rows
k	Operators evaluated per row in the filter	operators
c_{seq} , c_{tuple} , c_{op}	`seq_page_cost` , `cpu_tuple_cost` , `cpu_operator_cost`	cost units

From §11.10.2

FORMULA

Cardinality estimate under the independence assumption

$$\hat{r} = n \cdot \prod_{i=1}^{k} s_i, \qquad s_i \approx \text{freq}(v_i) \text{ or } \frac{1}{d_i}$$

Symbol	Meaning	Units
$\hat{r}$	Estimated rows satisfying all k predicates	rows
n	Rows in the table	rows
s_i	Selectivity of predicate i (fraction of rows passing it)	dimensionless
$\text{freq}(v_i)$	Sampled frequency of value v_i if it is a most-common value	dimensionless
d_i	Number of distinct values in the column	values

From §11.14.2

FORMULA

Time until transaction-ID wraparound

$$T_{\text{wrap}} = \frac{2^{31} - A}{\lambda_{\text{xid}}}$$

Symbol	Meaning	Units
T_{wrap}	Time until PostgreSQL must stop assigning transaction IDs	seconds
A	Current age of the oldest unfrozen transaction ID in the database	transactions
λ_{xid}	Rate of transactions that write (read-only transactions consume no ID)	transactions/second

From §11.15.3

FORMULA

Throughput ceiling of a single hot row

$$X_{\text{max}} = \frac{1}{t_{\text{hold}}}$$

Symbol	Meaning	Units
X_{max}	Maximum transactions per second that can update the same row	transactions/second
t_{hold}	Time each transaction holds the row's X lock (from its update to commit)	seconds

Chapter 12 — NoSQL Databases

From §12.4.5

FORMULA

DynamoDB capacity units

$$\text{WCU} = \sum_{\text{write types}} r_w \times \lceil \frac{s_w}{1\ \text{KB}} \rceil \times t_w \qquad \text{RCU} = \sum_{\text{read types}} r_r \times \lceil \frac{s_r}{4\ \text{KB}} \rceil \times c_r$$

Symbol	Meaning	Units
r_w, r_r	Rate of a given write or read type, including index writes	operations/second
s_w	Size of the item written (for a GSI, the projected size)	KB
s_r	Item size for GetItem; total returned size for Query	KB
t_w	1 for standard writes, 2 for transactional writes	dimensionless
c_r	0.5 eventually consistent, 1 strongly consistent, 2 transactional	dimensionless

From §12.4.5

FORMULA

On-demand versus provisioned break-even utilization

$$u^{*} = \frac{p_{\text{prov}}/3600}{p_{\text{od}}}$$

Symbol	Meaning	Units
u^{*}	Average utilization of provisioned capacity above which provisioned is cheaper	fraction (0–1)
p_{prov}	Price of one provisioned capacity unit for one hour	USD per unit-hour
p_{od}	Price of one on-demand request unit	USD per request unit

From §12.6.4

FORMULA

Cassandra partition cells and size

$$N_v = N_r \, (N_c - N_{pk} - N_s) + N_s$$

$$S_t = \sum_i \text{sizeOf}(c_{k_i}) + \sum_j \text{sizeOf}(c_{s_j}) + N_r \times \left(\sum_k \text{sizeOf}(c_{r_k}) + \sum_l \text{sizeOf}(c_{c_l}) \right) + N_v \times \text{sizeOf}(t_{\text{avg}})$$

Symbol	Meaning	Units
N_v	Number of cells (values) in the partition	cells
N_r	Number of rows in the partition	rows
N_c	Number of columns in the table	columns
N_{pk}	Number of partition-key columns	columns
N_s	Number of static columns	columns
S_t	Estimated partition size on disk, one replica, before compression	bytes
c_k, c_s, c_r, c_c	Partition-key, static, regular and clustering columns	bytes each
t_{avg}	Average per-cell metadata (the docs use the 8-byte timestamp)	bytes

From §12.7.3

FORMULA

Cost of a k-hop traversal

$$E_k = \sum_{i=1}^{k} d^i \qquad T_{\text{graph}} \approx E_k \times t_{\text{ptr}} \qquad T_{\text{join}} \approx E_k \times t_{\text{idx}}$$

Symbol	Meaning	Units
E_k	Edges expanded by a k-hop traversal without pruning	edges
d	Average out-degree (neighbors per node)	edges/node
k	Number of hops	hops
t_{ptr}	Cost of following one adjacency pointer	µs
t_{idx}	Cost of one index lookup on the edge table, growing with $\log N$	µs

From §12.8.3

FORMULA

Tiered time-series storage

$$S = N_{\text{series}} \times \sum_i \frac{R_i}{\Delta_i} \times b_i$$

Symbol	Meaning	Units
S	Total storage	bytes
N_{series}	Number of active series	series
R_i	Retention period of tier i	seconds
Δ_i	Sample (or rollup) interval of tier i	seconds
b_i	Bytes per stored point in tier i after compression	bytes/point

Chapter 13 — Storage Engine Internals

From §13.3.2

FORMULA

Records per slotted page

$$n = \lfloor \frac{P - H}{r + s} \rfloor$$

Symbol	Meaning	Units
n	Records that fit in one page	records
P	Page size	bytes
H	Page header size	bytes
r	Stored record size including per-record header and alignment padding	bytes
s	Slot (line pointer) size	bytes

From §13.4.3

FORMULA

Break-even reference interval (the “five-minute rule”)

$$T_{BE} = \frac{C_{dev} / IOPS_{dev}}{C_{RAM} \times P}$$

Symbol	Meaning	Units
T_{BE}	Break-even interval: cache a page if it is re-referenced more often than once per T_{BE}	seconds
C_{dev}	Price of one storage device	dollars
$IOPS_{dev}$	Random reads per second the device sustains	reads/s
C_{RAM}	Price of memory	dollars/byte
P	Page size	bytes

From §13.5.3

FORMULA

Redo work after a crash

$$W_{redo} \approx r_{WAL} \times \Delta t_{ckpt}, \qquad T_{redo} \approx \frac{W_{redo}}{b_{replay}}$$

Symbol	Meaning	Units
W_{redo}	Log volume between the redo start point and the crash	bytes
r_{WAL}	Rate at which the workload generates log	bytes/s
Δt_{ckpt}	Time between checkpoints (worst case, crash just before the next one)	s
T_{redo}	Duration of the redo pass	s
b_{replay}	Replay throughput, usually limited by random page reads, not log reads	bytes/s

From §13.8.1

FORMULA

Number of levels in a leveled LSM tree

$$L = \lceil \log_T \frac{N}{S_1} \rceil + 1 \quad (\text{levels } L_1 \ldots L_L, \text{ for } N > S_1)$$

Symbol	Meaning	Units
L	Number of levels below L0	levels
N	Total data size	bytes
S_1	Target size of level 1	bytes
T	Size ratio (fan-out) between adjacent levels	dimensionless

From §13.9.1

FORMULA

Approximate amplification of compaction strategies

$$WA_{level} \approx 2 + L\,\frac{T+1}{2}, \qquad WA_{tier} \approx 2 + L, \qquad WA_{lazy} \approx 2 + (L-1) + \frac{T+1}{2}$$

$$R_{level} \approx L, \qquad R_{tier} \approx (T-1)L, \qquad R_{lazy} \approx (T-1)(L-1) + 1$$

Symbol	Meaning	Units
WA	Write amplification: bytes written to storage per byte written by the application (the 2 counts the WAL and the memtable flush)	dimensionless
R	Sorted runs a point lookup may probe in the worst case (excluding memtables)	runs
L	Number of levels (tiers), $\lceil \log_T (N/M) \rceil$	levels
T	Size ratio between levels; runs merged at once in tiering	dimensionless
N	Data size	bytes
M	Memtable flush size	bytes

From §13.10

FORMULA

Bloom filter false-positive rate (summary of §17.4)

$$p \approx (1 - e^{-kn/m})^k, \qquad k_{opt} = \frac{m}{n} \ln 2, \qquad p_{opt} \approx 0.6185^{m/n}$$

Symbol	Meaning	Units
p	Probability that an absent key gets “probably yes”	dimensionless
m	Bits in the filter	bits
n	Keys inserted	keys
k	Hash functions	count

From §13.11.1

FORMULA

The three amplifications

$$RA = \frac{\text{bytes read from storage}}{\text{bytes returned}}, \qquad WA = \frac{\text{bytes written to storage}}{\text{bytes written by the application}}, \qquad SA = \frac{\text{bytes on storage}}{\text{bytes of live data}}$$

Symbol	Meaning	Units
RA	Read amplification (often counted instead as I/Os per lookup)	dimensionless
WA	Write amplification at the engine level, before the SSD’s own WAF	dimensionless
SA	Space amplification	dimensionless

From §13.13.2

FORMULA

Size of a dictionary-encoded, bit-packed column

$$S \approx \frac{N\lceil \log_2 d \rceil}{8} + d \cdot \bar{v}$$

Symbol	Meaning	Units
S	Encoded column size	bytes
N	Number of rows	rows
d	Number of distinct values	values
$\bar{v}$	Average size of one dictionary entry	bytes

Chapter 14 — Database Scaling: Replication, Partitioning and Sharding

From §14.3.1

FORMULA

Read capacity of a replicated group

$$R_{\max} = n \cdot (u \cdot c - w) \qquad \text{valid only while } w < u \cdot c$$

Symbol	Meaning	Units
$R_{\max}$	Maximum sustainable read rate across the group	reads/s
n	Number of nodes serving reads (replicas, plus the primary if it serves reads)	nodes
c	Capacity of one node, expressed in read-equivalent operations	ops/s
u	Target utilization, leaving head-room for spikes and failover	dimensionless (0–1)
w	Cost of applying the full write stream on one node, in read-equivalent operations	ops/s

From §14.4.1

FORMULA

Commit latency with k acknowledgments

$$T_{\text{commit}} \approx T_{\text{local}} + \text{RTT}_{(k)} + T_{\text{remote}}$$

Symbol	Meaning	Units
T_{commit}	Time from the client's `COMMIT` to the primary's reply	ms
T_{local}	Time to flush the log on the primary	ms
$\text{RTT}_{(k)}$	The k-th smallest round-trip time among the eligible replicas	ms
T_{remote}	Work the replica does before acknowledging (write, flush or apply)	ms
k	Number of replica acknowledgments required ($k = 0$ for asynchronous)	count

From §14.4.2

FORMULA

Replica catch-up time

$$T_{\text{catch-up}} = \frac{B}{r_{\text{apply}} - r_{\text{in}}} \qquad (r_{\text{apply}} > r_{\text{in}})$$

Symbol	Meaning	Units
$T_{\text{catch-up}}$	Time until the replica's lag returns to near zero	s
B	Backlog of unapplied changes when catch-up starts	transactions
r_{apply}	Rate at which the replica can apply changes	transactions/s
r_{in}	Rate at which the primary generates new changes during catch-up	transactions/s

From §14.7.4

FORMULA

Number of shards required

$$S = \left\lceil \max\left(\frac{D}{C_{\text{disk}} \cdot f},\ \frac{W}{w_{\text{node}} \cdot u},\ \frac{R}{n_{\text{read}} \cdot r_{\text{node}} \cdot u}\right)\right\rceil$$

Symbol	Meaning	Units
S	Number of shards (each one a replicated group)	shards
D	Data size at the planning horizon, one copy, including indexes	TB
C_{disk}	Usable storage per node	TB
f	Target fill fraction, leaving room for compaction, backups and growth between reshards	dimensionless (0–1)
W	Peak write rate at the horizon	writes/s
w_{node}	Sustainable write rate of one primary (from benchmarks)	writes/s
R	Peak read rate that reaches the database, after caches	reads/s
n_{read}	Nodes per shard that serve reads	nodes
r_{node}	Sustainable read rate of one node	reads/s
u	Target utilization at peak	dimensionless (0–1)

From §14.8.3

FORMULA

Expected keys moved on a membership change

$$E[\text{moved on add}] = \frac{K}{N+1} \qquad E[\text{moved on remove}] = \frac{K}{N}$$

Symbol	Meaning	Units
K	Total number of keys	keys
N	Number of nodes before the change	nodes

From §14.8.4

FORMULA

Load imbalance on a ring with virtual nodes

$$\mathrm{CV}(N,V) = \frac{\sqrt{\mathrm{Var}[L]}}{E[L]} = \sqrt{N-1}\,\sqrt{NV+1} \;\approx\; \frac{1}{\sqrt{V}} \quad (N \gg 1)$$

$$\frac{E[\max_i L_i]}{1/N} \approx 1 + z_N \cdot \mathrm{CV}(N,V) \quad (V \gtrsim 16), \qquad \frac{E[\max_i L_i]}{1/N} = H_N \quad (V = 1)$$

Symbol	Meaning	Units
L, L_i	Fraction of the key space owned by a node (by node i)	dimensionless (0–1)
N	Number of physical nodes	nodes
V	Virtual nodes per physical node	vnodes/node
CV	Coefficient of variation: standard deviation of a node's load divided by its mean	dimensionless
z_N	Expected maximum of N independent standard normal variables: 1.16 for $N = 5$, 1.54 for $N = 10$, 2.51 for $N = 100$, 3.24 for $N = 1,000$	dimensionless
H_N	N-th harmonic number $1 + \frac{1}{2} + \ldots + \frac{1}{N} \approx \ln N + 0.577$	dimensionless

From §14.10.1

FORMULA

Salt factor for a hot key

$$s = \left\lceil \frac{\lambda_{\text{key}}}{u \cdot c_{\text{part}}} \right\rceil \qquad \text{read amplification} = s$$

Symbol	Meaning	Units
s	Number of sub-keys (salt factor)	sub-keys
λ_{key}	Write rate to the hot key	writes/s
c_{part}	Write capacity of one partition	writes/s
u	Target utilization of a partition	dimensionless (0–1)

From §14.13.1

FORMULA

Server connections needed behind a transaction pooler (Little's Law)

$$C_{\text{server}} = \lambda_{\text{tx}} \cdot W_{\text{tx}} \cdot (1 + h)$$

Symbol	Meaning	Units
C_{server}	Server connections the pool needs	connections
λ_{tx}	Transaction rate through the pooler	transactions/s
W_{tx}	Time a transaction holds a server connection, from `BEGIN` to `COMMIT`	s
h	Headroom fraction for bursts	dimensionless

Chapter 15 — Object, Block and File Storage

From §15.3.1

FORMULA

Access time for one read

$$t \approx t_{\text{first}} + \frac{S}{B} \qquad t_{\text{parallel}} \approx t_{\text{first}} + \frac{S}{\min\left(c \cdot B,\ B_{\text{NIC}}\right)}$$

Symbol	Meaning	Units
t	Time to read the whole request	s
t_{first}	Time to first byte (network round trip, lookup, seek or flash read)	s
S	Bytes requested	bytes
B	Throughput of one stream (connection or volume)	bytes/s
c	Number of parallel streams (for example byte-range GETs)	count
B_{NIC}	Client network interface limit	bytes/s

From §15.4.2

FORMULA

Prefixes needed for a target request rate

$$N_{\text{prefix}} = \lceil \frac{R_{\text{peak}}}{r_{\text{prefix}}} \rceil$$

Symbol	Meaning	Units
N_{prefix}	Distinct key prefixes over which requests must be spread	count
R_{peak}	Peak request rate of one type (writes or reads)	requests/s
r_{prefix}	Sustained rate one partitioned prefix supports	requests/s

From §15.5.1

FORMULA

Metadata memory for an in-memory namespace

$$M = N_{\text{files}} \cdot b_{\text{file}} + N_{\text{chunks}} \cdot b_{\text{chunk}} \qquad N_{\text{chunks}} \approx \sum_{\text{files}} \lceil \frac{\text{file size}}{C} \rceil$$

Symbol	Meaning	Units
M	Memory needed by the metadata server	bytes
N_{files}	Number of files (and directories)	count
N_{chunks}	Number of chunks or blocks	count
$b_{\text{file}}, b_{\text{chunk}}$	Bytes of metadata per file and per chunk	bytes
C	Chunk (block) size	bytes

From §15.6.1

FORMULA

Overhead, fault tolerance and repair reads of a (k, m) code

$$\text{overhead} = \frac{k+m}{k} \qquad \text{tolerated losses} = m \qquad \text{repair read per lost fragment} = r \times F$$

Symbol	Meaning	Units
k	Data fragments per stripe	count
m	Parity fragments per stripe	count
overhead	Raw bytes stored per logical byte	dimensionless (×)
r	Fragments read to rebuild one: 1 for replication, k for RS, local group size for LRC	count
F	Fragment size	bytes

From §15.6.1

FORMULA

Approximate annual probability of losing a stripe (generalizing §4.16)

$$P_{\text{loss/yr}} \approx \frac{n!}{(n-m-1)!}\,\lambda^{m+1}\,T^{m} \times H, \qquad n = k+m, \quad \lambda T \ll 1$$

Symbol	Meaning	Units
n	Fragments per stripe (replicas for replication)	count
m	Losses tolerated	count
λ	Per-fragment (per-disk) failure rate, AFR/8,760	failures/hour
T	Mean time to detect and rebuild one lost fragment	hours
H	Hours per year (8,760)	hours/year

From §15.6.1

FORMULA

Daily repair read traffic in a fleet

$$R_{\text{day}} = \frac{N_{\text{disk}} \times \text{AFR}}{365} \times D_{\text{used}} \times r$$

Symbol	Meaning	Units
R_{day}	Bytes read per day to rebuild failed disks	bytes/day
N_{disk}	Disks in the fleet	count
AFR	Annualized failure rate	fraction/year
D_{used}	Data stored per disk	bytes
r	Fragments read per fragment rebuilt	count

From §15.8.1

FORMULA

Part sizing and upload time

$$p_{\text{min}} = \max\left(5\ \text{MiB}, \frac{S}{N_{\text{max}}}\right) \qquad N = \left\lceil \frac{S}{p} \right\rceil \qquad t \approx \frac{S}{\min\left(c \cdot b,\ B_{\text{NIC}}\right)}$$

Symbol	Meaning	Units
S	Object size	bytes
p, p_{min}	Chosen and minimum part size	bytes
N_{max}	Maximum parts per upload (10,000 in S3)	count
N	Number of parts	count
c	Concurrent part uploads	count
b	Throughput per connection	bytes/s
B_{NIC}	Client network limit	bytes/s

From §15.8.3

FORMULA

Expected chunk size and deduplication index size

$$E[\text{chunk}] \approx s_{\min} + 2^{b} \qquad M_{\text{index}} = \frac{U}{E[\text{chunk}]} \times e$$

Symbol	Meaning	Units
b	Number of hash bits that must match (boundary probability 2^{-b} per byte)	bits
$s_{\min}$	Minimum chunk size (no boundary is accepted earlier)	bytes
$E[\text{chunk}]$	Expected chunk size (ignoring the maximum cap)	bytes
U	Unique data stored	bytes
e	Index entry size (hash plus location)	bytes

From §15.9

FORMULA

Cost of one cohort of data over its lifetime

$$C = \sum_{\text{tiers } i} V \cdot p_i \cdot t_i \; + \sum_{\text{transitions } j} N_{\text{obj}} \cdot q_j \; + \; V_{\text{read}} \cdot g$$

Symbol	Meaning	Units
V	Volume of the cohort	GB
p_i, t_i	Storage price and months spent in tier i	$/GB-month, months
N_{obj}	Number of objects	count
q_j	Price per lifecycle transition request into tier j	$/request
V_{read}, g	Data retrieved and its per-GB retrieval fee	GB, $/GB

Chapter 16 — Search Systems

From §16.3.4

FORMULA

Variable-byte length of an integer

$$\text{bytes}(g) = \max\left(1, \left\lceil \frac{\lfloor \log_2 g \rfloor + 1}{7} \right\rceil\right)$$

Symbol	Meaning	Units
g	The integer to encode, here a docID gap ($g \geq 1$)	count
$\lfloor \log_2 g \rfloor + 1$	Number of significant bits in g	bits
$\text{bytes}(g)$	Encoded length, 7 payload bits per byte	bytes

From §16.3.5

FORMULA

Inverted index size (one copy)

$$S_{\text{index}} \approx \underbrace{N \cdot u \cdot (b_d + b_f)}_{\text{docIDs + frequencies}} + \underbrace{N \cdot L \cdot b_p}_{\text{positions}} + \underbrace{M \cdot b_t}_{\text{dictionary}}$$

Symbol	Meaning	Units
N	Number of documents	documents
u	Average number of *distinct* terms per document (postings per document)	postings/document
L	Average number of tokens per document (one position each)	tokens/document
b_d, b_f	Compressed bytes per docID gap and per term frequency	bytes
b_p	Compressed bytes per position gap	bytes
M	Number of distinct terms in the collection	terms
b_t	Dictionary bytes per term (term text + pointers + statistics)	bytes

From §16.5.2

FORMULA

TF-IDF weight of term t in document d

$$w_{t,d} = \mathrm{tf}_{t,d} \times \log_{10} \frac{N}{\mathrm{df}_t} \qquad \mathrm{score}(q,d) = \sum_{t \in q} w_{t,d}$$

Symbol	Meaning	Units
$\mathrm{tf}_{t,d}$	Occurrences of term t in document d	count
N	Documents in the collection	documents
df_t	Documents containing t	documents
$w_{t,d}$	Weight of t in d	dimensionless

From §16.5.3

FORMULA

BM25 inverse document frequency (Lucene form)

$$\mathrm{IDF}(t) = \ln\left(1 + \frac{N - n_t + 0.5}{n_t + 0.5}\right)$$

Symbol	Meaning	Units
N	Documents in the collection (or shard; see §16.7.5)	documents
n_t	Documents containing term t (document frequency)	documents
$\mathrm{IDF}(t)$	Weight of a match on t	dimensionless

From §16.5.3

FORMULA

BM25 score

$$\text{score}(q,d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f_{t,d}\,(k_1 + 1)}{f_{t,d} + k_1\left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)}$$

Symbol	Meaning	Units
$f_{t,d}$	Frequency of term t in document (field) d	count
$\lvert d\rvert$	Length of d in terms after analysis	terms
avgdl	Average length of that field across the collection	terms
k_1	Term-frequency saturation parameter; Lucene default 1.2 (typical 1.2–2.0)	dimensionless
b	Length-normalization strength, 0 (none) to 1 (full); default 0.75	dimensionless
$\text{score}(q,d)$	Relevance score; only comparable within one query	dimensionless

From §16.6.3

FORMULA

Write amplification of tiered merging

$$W \approx 1 + \log_m \frac{S_{\text{max}}}{S_{\text{flush}}}$$

Symbol	Meaning	Units
W	Bytes written to disk per byte of new index data	dimensionless
m	Merge factor: segments merged together per tier	count
S_{max}	Maximum merged segment size	bytes
S_{flush}	Typical size of a newly flushed segment	bytes

From §16.7.3

FORMULA

Scatter-gather tail latency

$$P(\text{query fast}) = p^S \qquad q_{\text{shard}} = q_{\text{query}}^{1/S}$$

Symbol	Meaning	Units
p	Probability one shard answers within threshold T	dimensionless
S	Number of shards the query waits for	count
$P(\text{query fast})$	Probability all S shards answer within T	dimensionless
q_{query}	Target percentile for the whole query (e.g. 0.99)	dimensionless
q_{shard}	Percentile each shard must meet at T	dimensionless

From §16.7.4

FORMULA

Deep pagination cost at the coordinator

$$E = S \cdot (\mathit{from} + \mathit{size})$$

Symbol	Meaning	Units
S	Shards queried	count
from	Offset of the first requested result	results
size	Results per page	results
E	Entries every query phase must return and the coordinator must merge	entries

From §16.9.1

FORMULA

Autocomplete load and precomputed-index memory

$$\lambda_{ac} = \lambda_{search} \cdot c \cdot f \qquad M_{top\text{-}k} \approx P \cdot k \cdot b$$

Symbol	Meaning	Units
λ_{search}	Searches started per second	searches/s
c	Characters typed per search before selecting or submitting	characters
f	Fraction of keystrokes that send a request (after debounce and client caching)	dimensionless
λ_{ac}	Suggestion request rate	requests/s
P	Number of trie nodes (distinct prefixes) with stored suggestions	nodes
k	Suggestions stored per prefix	count
b	Bytes per stored suggestion reference	bytes
$M_{top\text{-}k}$	Memory for the precomputed lists	bytes

From §16.10.3

FORMULA

End-to-end freshness latency

$$T_{\text{fresh}} \approx T_{\text{cdc}} + T_{\text{queue}} + T_{\text{batch}} + T_{\text{index}} + T_{\text{refresh}}$$

Symbol	Meaning	Units
T_{cdc}	Commit to event published in Kafka	s
T_{queue}	Time the event waits for the indexer (consumer lag)	s
T_{batch}	Wait for a bulk request to fill or its timer to fire	s
T_{index}	Bulk request execution on primary and replicas	s
T_{refresh}	Wait for the next refresh: uniform in $[0, I]$, mean $I/2$	s
T_{fresh}	Commit in source database to searchable	s

Chapter 17 — Data Structures and Algorithms for Scale

From §17.3.3

FORMULA

Expected bucket load and the chi-square uniformity test

$$\mu = \frac{n}{B}, \qquad \sigma \approx \sqrt{\frac{n}{B}}, \qquad \chi^2 = \sum_{i=1}^{B} \frac{(c_i - \mu)^2}{\mu}, \qquad \mathbb{E}[\chi^2] \approx B - 1, \quad \operatorname{sd}(\chi^2) \approx \sqrt{2(B-1)}$$

Symbol	Meaning	Units
n	Number of keys hashed	keys
B	Number of buckets	buckets
μ	Expected keys per bucket	keys/bucket
σ	Standard deviation of a bucket's count under a uniform hash	keys/bucket
c_i	Observed number of keys in bucket i	keys
χ^2	Chi-square statistic: total squared deviation from uniform	dimensionless

From §17.4.3

FORMULA

Bloom filter false-positive rate

$$p \approx \left(1 - e^{-kn/m}\right)^{k}$$

Symbol	Meaning	Units
p	Probability that a query for an item not in the set returns “probably present”	probability (0–1)
n	Number of items inserted	items
m	Size of the bit array	bits
k	Number of hash functions (bit positions per item)	count
$e^{-kn/m}$	Approximate fraction of bits still 0	fraction

From §17.4.3

FORMULA

Optimal number of hash functions

$$k^{*} = \frac{m}{n} \ln 2 \approx 0.693 \frac{m}{n}, \qquad p\big|_{k = k^{*}} = \left(\tfrac{1}{2}\right)^{k^{*}}$$

Symbol	Meaning	Units
k^{*}	Number of hash functions that minimizes p (round to an integer)	count
m/n	Bits per element	bits/item
$\ln 2$	Natural logarithm of 2, ≈ 0.693	dimensionless

From §17.4.3

FORMULA

Bits per element for a target false-positive rate

$$\frac{m}{n} = \frac{-\ln p}{(\ln 2)^2} \approx 1.44 \log_2 \frac{1}{p}, \qquad k = -\log_2 p$$

Symbol	Meaning	Units
m/n	Bits of filter needed per inserted item	bits/item
p	Target false-positive rate	probability
$(\ln 2)^2$	≈ 0.4805	dimensionless
k	Matching number of hash functions (rounded)	count

From §17.4.7

FORMULA

Cuckoo filter false-positive bound and space cost

$$\varepsilon \le \frac{2b}{2^f}, \qquad f \ge \log_2 \frac{2b}{\varepsilon}, \qquad \text{bits per item} = \frac{f}{\alpha} \approx \frac{\log_2 (1/\varepsilon) + \log_2 (2b)}{\alpha}$$

Symbol	Meaning	Units
ε	False-positive rate	probability
b	Slots per bucket (4 is the common choice)	slots
f	Fingerprint length	bits
α	Load factor reached before inserts fail (≈ 0.95 for $b = 4$)	fraction

From §17.5.1

FORMULA

HyperLogLog estimate and standard error

$$E = \alpha_m m^2 \left(\sum_{j=1}^{m} 2^{-M[j]} \right)^{-1}, \qquad \alpha_m \approx \frac{0.7213}{1 + 1.079/m}, \qquad \text{SE} \approx \frac{1.04}{\sqrt{m}}, \qquad \text{memory} = m \times 6 \text{ bits}$$

Symbol	Meaning	Units
E	Estimated number of distinct items	items
m	Number of registers, $m = 2^p$	registers
$M[j]$	Value of register j: the largest first-1-bit position seen	count
α_m	Bias-correction constant (valid for $m \geq 128$)	dimensionless
SE	Relative standard error of E	fraction
6 bits	Register width: enough to hold values up to 63, sufficient for a 64-bit hash	bits

From §17.6.1

FORMULA

Count-Min sizing and error bound

$$w = \lceil \frac{e}{\varepsilon} \rceil, \qquad d = \lceil \ln \frac{1}{\delta} \rceil, \qquad f_x \leq \hat{f}_x \leq f_x + \varepsilon N \text{ with probability } \geq 1 - \delta$$

Symbol	Meaning	Units
w	Counters per row (width)	counters
d	Rows, one hash function each (depth)	rows
ε	Error as a fraction of the stream total	fraction
δ	Probability that the error bound is exceeded for a given query	probability
N	Sum of all updates in the stream	events
f_x, $\hat{f}_x$	True and estimated count of item x	events
e	Euler's number, ≈ 2.718	dimensionless

From §17.6.2

FORMULA

Space-Saving guarantee

$$c_x - f_x \le c_{\min} \le \frac{N}{k}, \qquad f_x > \frac{N}{k} \Rightarrow x \text{ is monitored}$$

Symbol	Meaning	Units
k	Number of counters	counters
N	Stream length	events
c_x, f_x	Reported and true count of monitored item x	events
$c_{\min}$	Smallest counter value	events

From §17.7.1

FORMULA

Skip list levels, search cost and memory

$$L(n) = \log_{1/p} n, \qquad \mathbb{E}[\text{search path}] \le \frac{L(n)}{p} + \frac{1}{1-p}, \qquad \mathbb{E}[\text{pointers per node}] = \frac{1}{1-p}$$

Symbol	Meaning	Units
n	Number of elements	elements
p	Probability of promoting a node to the next level	probability
$L(n)$	Expected number of levels in use	levels
Search path	Forward moves plus level drops during one search	steps
Pointers per node	Average forward pointers stored per element	pointers

From §17.8.1

FORMULA

Merkle proof size and anti-entropy cost

$$h = \lceil \log_2 n \rceil, \qquad \text{proof size} = h \times H_{\text{bytes}}, \qquad \text{hashes compared} \le 1 + 2\,k\,h$$

Symbol	Meaning	Units
n	Number of leaves (records or key ranges)	leaves
h	Tree height	levels
H_{bytes}	Size of one hash (32 for SHA-256)	bytes
k	Number of divergent leaves between two replicas	leaves

From §17.9.2

FORMULA

Geohash cell size

$$b_{\text{lon}} = \lceil \frac{5L}{2} \rceil, \quad b_{\text{lat}} = \lfloor \frac{5L}{2} \rfloor, \quad W = \frac{360^\circ}{2^{b_{\text{lon}}}} \times 111.32 \cos\varphi \text{ km}, \quad H = \frac{180^\circ}{2^{b_{\text{lat}}}} \times 110.57 \text{ km}$$

Symbol	Meaning	Units
L	Geohash length	characters
b_{lon}, b_{lat}	Bits spent on longitude and latitude	bits
W, H	Cell width (east–west) and height (north–south)	km
φ	Latitude of the cell	degrees
111.32, 110.57	Approximate km per degree of longitude at the equator and of latitude	km/degree

From §17.9.4

FORMULA

Average S2 cell area at level L

$$\bar{A}_L \approx \frac{4\pi R^2}{6 \times 4^L} = \frac{510.1 \times 10^6 \text{ km}^2}{6 \times 4^L}$$

Symbol	Meaning	Units
$\bar{A}_L$	Average area of a level-L cell	km²
R	Earth's mean radius, ≈ 6,371 km	km
L	Cell level, 0 (a cube face) to 30	level

From §17.10.2

FORMULA

Birthday bound for random identifiers

$$P_{\text{collision}} \approx 1 - e^{-n^2/(2N)} \approx \frac{n^2}{2N} \text{ (when small)}, \qquad n_{50\%} \approx \sqrt{2N \ln 2} \approx 1.18\sqrt{N}$$

Symbol	Meaning	Units
n	Number of IDs generated within one collision domain	IDs
N	Number of possible values, $2^{\text{random bits}}$	values
$P_{\text{collision}}$	Probability that at least two IDs are equal	probability
$n_{50\%}$	Number of IDs at which a collision becomes a coin flip	IDs

From §17.10.3

FORMULA

Snowflake capacity and lifetime

$$\text{rate}_{\text{machine}} = 2^{S} \times 1000 \text{ IDs/s}, \qquad \text{rate}_{\text{cluster}} = 2^{M} \times \text{rate}_{\text{machine}}, \qquad \text{lifetime} = \frac{2^{T}}{1000 \times 3600 \times 24 \times 365.25} \text{ yea}$$

Symbol	Meaning	Units
T	Timestamp bits (millisecond resolution)	bits
M	Machine-ID bits	bits
S	Sequence bits	bits
rate	Maximum sustained ID generation rate	IDs/s
lifetime	Time from the epoch until the timestamp field overflows	years

From §17.12.1

FORMULA

Reservoir sampling inclusion probability

$$P(\text{item } i \text{ in sample after } n \text{ items}) = \frac{k}{i} \times \prod_{j=i+1}^{n} \left(1 - \frac{1}{j}\right) = \frac{k}{i} \times \frac{i}{n} = \frac{k}{n}, \qquad \mathbb{E}[\text{replacements}] \approx k \ln \frac{n}{k}$$

Symbol	Meaning	Units
k	Reservoir size	items
n	Items seen so far	items
i	Arrival position of a particular item ($i > k$)	position
$1 - 1/j$	Probability that item j does not evict this item: it is chosen with probability k/j and then evicts one of k slots	probability

Chapter 18 — Scalability

From §18.3.1

FORMULA

Relative cost per unit of throughput on one node

$$k(N) = \frac{N}{C(N)} = 1 + \alpha(N - 1) + \beta N(N - 1)$$

Symbol	Meaning	Units
N	Cores (or vCPUs) in the node; price assumed proportional to N	count
$C(N)$	USL relative capacity: throughput at N cores divided by throughput at 1 core	dimensionless
α	Contention coefficient inside the node	dimensionless
β	Coherency coefficient inside the node	dimensionless
$k(N)$	Cost per unit of throughput relative to a 1-core node	dimensionless (×)

From §18.7.1

FORMULA

Marginal throughput of the N-th node

$$\Delta X(N) = X_1 [C(N) - C(N - 1)] \qquad E(N) = \frac{C(N)}{N}$$

Symbol	Meaning	Units
X_1	Throughput of one node (measured)	requests/s
$C(N)$	USL relative capacity at N nodes (§4.13)	dimensionless
$\Delta X(N)$	Extra throughput gained by adding node N	requests/s
$E(N)$	Scaling efficiency: fraction of linear scaling achieved	dimensionless (0–1)

From §18.8.3

FORMULA

Target-tracking replica count (Kubernetes HPA)

$$n_{desired} = \lceil n_{current} \times \frac{m_{current}}{m_{target}} \rceil \quad \text{if } |\frac{m_{current}}{m_{target}} - 1| > \tau$$

Symbol	Meaning	Units
$n_{current}$	Current number of replicas	pods
$n_{desired}$	Replica count the autoscaler requests, clamped to [min, max]	pods
$m_{current}$	Current per-replica metric, averaged over replicas	metric units (e.g. % CPU, RPS)
m_{target}	Target per-replica metric	same units
τ	Tolerance band within which no action is taken (default 0.1 as of 2026)	dimensionless

From §18.8.4

FORMULA

Target utilization that survives reaction time

$$u_{target} \leq \frac{u_{safe}}{1 + r \times T_{react}}$$

Symbol	Meaning	Units
u_{target}	Utilization set point for the autoscaler	fraction
u_{safe}	Highest utilization at which latency still meets the objective (from load tests and §4.10)	fraction
r	Fastest expected load growth, relative to current load	fraction per minute
T_{react}	Reaction time: metric delay + evaluation + provisioning + start + warm-up	minutes

From §18.8.5

FORMULA

Connection budget

$$n_{max} \times p + \sum_{j} n_j p_j \le M - R$$

Symbol	Meaning	Units
n_{max}	Maximum replicas of the scaled service (autoscaler `maxReplicas`)	pods
p	Maximum pool size per replica	connections
n_j, p_j	Replicas and pool size of every other client service j	pods, connections
M	Database `max_connections`	connections
R	Connections reserved for superusers, replication, monitoring and migrations	connections

From §18.11.2

FORMULA

Provisioned capacity with headroom

$$N = \lceil \frac{R_{peak}\,(1+g)^{L}}{c \times u_{target}} \rceil \qquad N_{total} = N + k \quad \text{or} \quad N_{total} = \lceil N \times \frac{Z}{Z-f} \rceil$$

Symbol	Meaning	Units
R_{peak}	Current design peak load	requests/s
g	Demand growth rate per period	fraction per month
L	Lead time to add capacity	months
c	Per-node throughput at which latency meets the objective (measured)	requests/s per node
u_{target}	Target utilization (queueing and reaction-time headroom, §18.8.4)	fraction
N	Nodes needed to serve the load with headroom	nodes
k	Spare nodes for failures and deploys (N+1: $k = 1$; N+2: $k = 2$)	nodes
Z, f	Failure domains (zones) used, and how many may fail at once	count
N_{total}	Nodes to provision	nodes

From §18.11.3

FORMULA

Peak-to-average ratio and average fleet utilization

$$PAR = \frac{R_{peak}}{R_{avg}} \qquad \bar{u} = \frac{R_{avg}}{N_{total} \times c}$$

Symbol	Meaning	Units
R_{avg}	Average load over a day (or week)	requests/s
PAR	Peak-to-average ratio	dimensionless
$\bar{u}$	Average utilization of a statically provisioned fleet	fraction

From §18.11.4

FORMULA

Time until a component's capacity is exhausted

$$D(t) = D_0(1+g)^t \qquad t_{exhaust} = \frac{\ln(C_{usable}/D_0)}{\ln(1+g)} \qquad t_{start} = t_{exhaust} - L_{fix}$$

Symbol	Meaning	Units
D_0	Current peak demand on the component	operations/s
g	Compound growth rate per period	fraction per month
C_{usable}	Usable capacity: limit × target utilization	operations/s
$t_{exhaust}$	Periods until demand reaches usable capacity	months
L_{fix}	Lead time of the fix, including a safety margin	months
t_{start}	Latest month to start the fix	months

Chapter 19 — Load Balancing

From §19.3.1

FORMULA

Balancer bandwidth by forwarding mode

$$B_{\text{proxy}} = R(s_{\text{req}} + s_{\text{resp}}), \qquad B_{\text{DSR}} = R\,s_{\text{req}}$$

Symbol	Meaning	Units
B_{proxy}	Bandwidth through the balancer when both directions cross it	bytes/second
B_{DSR}	Bandwidth through the balancer with direct server return	bytes/second
R	Request rate	requests/second
s_{req}, s_{resp}	Average request and response size on the wire	bytes

From §19.4.1

FORMULA

Weighted round-robin share

$$\lambda_i = \lambda \cdot \frac{w_i}{\sum_{j=1}^{N} w_j}, \qquad \rho_i = \frac{\lambda_i}{\mu_i}$$

Symbol	Meaning	Units
λ	Total request rate into the pool	requests/second
λ_i	Request rate sent to backend i	requests/second
w_i	Configured weight of backend i	dimensionless
N	Number of backends	count
μ_i	Capacity (service rate) of backend i	requests/second
ρ_i	Utilization of backend i	dimensionless

From §19.4.2

FORMULA

Little's Law per backend (§4.11)

$$L_i = \lambda_i W_i \quad \Rightarrow \quad \lambda_i = \frac{L_i}{W_i}$$

Symbol	Meaning	Units
L_i	Average in-flight requests at backend i (its "connections")	requests
λ_i	Request rate sent to backend i	requests/second
W_i	Average time a request spends at backend i	seconds

From §19.4.3

FORMULA

Exponentially weighted moving average

$$S_k = \alpha x_k + (1-\alpha) S_{k-1}, \qquad S_k - x^* = (1-\alpha)^k (S_0 - x^*) \text{ after a step to } x^*, \qquad \alpha = 1 - e^{-\Delta t/\tau}$$

Symbol	Meaning	Units
S_k	Latency estimate after k samples	ms
x_k	The k-th observed latency	ms
α	Smoothing factor, $0 < \alpha \leq 1$	dimensionless
x^*	New steady latency after a sudden change	ms
Δt, τ	Time since previous sample; decay time constant (time-based variant)	seconds

From §19.4.4

FORMULA

Maximum load with one and with d random choices [Azar et al. 1999]

$$\text{one choice: } \max \text{ load} = \frac{\ln n}{\ln \ln n}(1 + o(1)), \qquad d \geq 2 \text{ choices: } \max \text{ load} = \frac{\ln \ln n}{\ln d} + \Theta(1) \quad \text{(with high pr}$$

Symbol	Meaning	Units
n	Number of balls (requests) and of bins (servers)	count
d	Number of random bins examined per ball	count
$\mathrm{maxload}$	Largest number of balls in any bin	balls
$o(1)$, $\Theta(1)$	A term that vanishes as n grows; a bounded constant	—

From §19.4.4

FORMULA

Queue-length tail in the supermarket model (large n)

$$s_i = \lambda^i \quad (d = 1), \qquad s_i = \lambda^{\frac{d^i - 1}{d - 1}} \quad (d \geq 2), \qquad \text{so for } d = 2\text{: } s_i = \lambda^{2^i - 1}$$

Symbol	Meaning	Units
s_i	Fraction of servers with at least i requests (queued or in service)	dimensionless
λ	Utilization: arrival rate per server divided by service rate	dimensionless
d	Number of queues sampled per request	count
i	Queue-length threshold	requests

From §19.4.7

FORMULA

Bounded-load capacity

$$\text{cap} = \lceil (1 + \varepsilon) \frac{m}{n} \rceil$$

Symbol	Meaning	Units
cap	Maximum items any backend may hold	items
ε	Allowed overload above the average, $\varepsilon > 0$	dimensionless
m	Items currently assigned (for request routing: in-flight requests)	items
n	Number of backends	count

From §19.4.8

FORMULA

Maglev preference permutation

$$\text{offset}_i = h_1(\text{name}_i) \bmod M, \qquad \text{skip}_i = h_2(\text{name}_i) \bmod (M-1) + 1, \qquad \text{perm}_i[j] = (\text{offset}_i + j \cdot \text{skip}_i) \text{ ı}$$

Symbol	Meaning	Units
M	Lookup-table size, a prime	entries
h_1, h_2	Two independent hash functions	—
offset_i	Backend i's first preferred position	entry index
skip_i	Step between backend i's successive preferences, $1 \le \text{skip}_i \le M-1$	entries
$\text{perm}_i[j]$	Backend i's j-th preferred position	entry index

From §19.5.1

FORMULA

Detection time and requests lost to a dead backend

$$T_{\text{detect}} \approx F \cdot I + T_o, \qquad N_{\text{lost}} \approx \frac{\lambda}{n} T_{\text{detect}}$$

Symbol	Meaning	Units
T_{detect}	Worst-case time from failure to removal	seconds
F	Consecutive failures required (fall threshold)	count
I	Probe interval	seconds
T_o	Probe timeout	seconds
N_{lost}	Requests sent to the dead backend meanwhile (without retries)	requests
λ, n	Total request rate; number of backends	requests/second; count

From §19.7

FORMULA

Safe utilization for active-active with k tolerated failures

$$U_{\max} = \frac{N - k}{N}$$

Symbol	Meaning	Units
$U_{\max}$	Highest normal utilization per balancer that still fits all traffic after k failures	dimensionless
N	Balancers in the active–active group	count
k	Simultaneous failures to tolerate	count

From §19.8.1

FORMULA

VRRP failover time [RFC 5798, §6.1]

$$T_{\text{skew}} = \frac{(256 - P)\,A}{256}, \qquad T_{\text{down}} = 3A + T_{\text{skew}}$$

Symbol	Meaning	Units
A	Advertisement interval (VRRPv3 allows centisecond values)	seconds
P	Backup router's priority, 1–254	dimensionless
T_{skew}	Extra wait so higher-priority backups take over first	seconds
T_{down}	Master-down interval: silence before a backup takes over	seconds

From §19.8.3

FORMULA

Flows remapped when one of N next hops is removed

$$f_{\text{mod}} \approx 1 - \frac{1}{N}, \qquad \tfrac{1}{4} \lesssim f_{\text{threshold}} \leq \tfrac{1}{2}, \qquad f_{\text{min}} = \frac{1}{N}$$

Symbol	Meaning	Units
f_{mod}	Fraction of flows that change next hop with `hash mod N`	dimensionless
$f_{\text{threshold}}$	Same, for the hash-threshold method analyzed in RFC 2992 (depends on which region is removed)	dimensionless
f_{min}	Minimum: only the failed hop's flows move (resilient or consistent hashing)	dimensionless
N	Next hops (balancers) before the failure	count

Chapter 20 — Caching

From §20.3.2

FORMULA

Effective latency and backend load of a look-aside cache

$$T_{\text{eff}} = t_c + (1 - h)\, t_b \qquad R_{\text{db}} = (1 - h)\, R_{\text{read}} + R_{\text{write}}$$

Symbol	Meaning	Units
T_{eff}	Mean latency of a read through the cache	ms
t_c	Cache lookup latency (paid on hits and misses)	ms
t_b	Backend read latency, including the fill	ms
h	Cache hit ratio	dimensionless (0–1)
R_{read}	Read requests reaching the cache	requests/s
R_{write}	Writes, which go to the database regardless	requests/s
R_{db}	Requests reaching the database	requests/s

From §20.3.3

FORMULA

Zipf request probability and the hit ratio of an ideal top-C cache

$$p_i = \frac{1/i^s}{H_{N,s}}, \qquad H_{n,s} = \sum_{j=1}^{n} \frac{1}{j^s}, \qquad h_{\text{ideal}}(C) = \frac{H_{C,s}}{H_{N,s}} \overset{s=1}{\approx} \frac{\ln C + \gamma}{\ln N + \gamma}$$

Symbol	Meaning	Units
N	Number of distinct keys	keys
C	Cache capacity	entries
i	Popularity rank (1 = most popular)	dimensionless
s	Zipf skew exponent	dimensionless
p_i	Probability that a request is for rank i	dimensionless
$H_{n,s}$	Generalized harmonic number	dimensionless
$h_{\text{ideal}}(C)$	Hit ratio of a cache that always holds the top C keys	dimensionless
γ	Euler–Mascheroni constant, ≈ 0.5772	dimensionless

From §20.6.5

FORMULA

Slab internal fragmentation

$$c_{k+1} \approx f \cdot c_k, \qquad \text{waste}(x) = c_k - x \text{ for } c_{k-1} < x \le c_k, \qquad \frac{\text{waste}}{c_k} < 1 - \frac{1}{f}, \qquad \mathbb{E}[\frac{\text{waste}}{c_k}] \approx \frac{1}{2}(1 - \frac{1}{f})$$

Symbol	Meaning	Units
c_k	Chunk size of slab class k (rounded to 8 bytes)	bytes
f	Growth factor between classes (`-f`)	dimensionless
x	Item size, including header	bytes
waste	Unused bytes in the chunk	bytes

From §20.7.5

FORMULA

Data at risk under write-back

$$W_{\text{risk}} \approx r_w \times T_{\text{flush}}, \qquad \text{write reduction} = \frac{r_w}{r_{\text{flush}}}$$

Symbol	Meaning	Units
W_{risk}	Acknowledged writes that a crash can lose	writes
r_w	Write rate into the cache	writes/s
T_{flush}	Maximum time a dirty entry waits before flushing	s
r_{flush}	Database writes per second after coalescing	writes/s

From §20.8

FORMULA

TTL with uniform jitter

$$T_i = T_{\text{base}} + U_i, \quad U_i \sim \text{Uniform}(0, J), \qquad \text{peak refills/s} \approx \frac{K}{\max(J, t_{\text{write}})}$$

Symbol	Meaning	Units
T_i	TTL given to key i	s
T_{base}	Base TTL	s
J	Jitter range	s
K	Keys written together	keys
t_{write}	Time span over which the keys were originally written	s

From §20.11.2

FORMULA

XFetch early-recomputation rule and its consequences (Poisson arrivals)

$$\text{recompute if } \; t - \delta\beta \, \ln(u) \geq t_{\text{exp}}, \quad u \sim \text{Uniform}(0,1] \qquad \Rightarrow \qquad P(\text{early} \mid r) = e^{-r/(\delta\beta)}$$

$$\mathbb{E}[\text{lead}] \approx \delta\beta(\ln(\lambda\delta\beta) + \gamma), \qquad \mathbb{E}[\text{recomputes per refresh}] \approx e^{1/\beta}$$

Symbol	Meaning	Units
t	Current time	s
t_{exp}	Entry's expiry time	s
$r = t_{\text{exp}} - t$	Time remaining before expiry	s
δ	Measured recompute time, stored with the value	s
β	Aggressiveness; 1 is the recommended default	dimensionless
u	Uniform random number per request	dimensionless
λ	Request rate for the key	requests/s
γ	Euler–Mascheroni constant, ≈ 0.5772	dimensionless

From §20.15

FORMULA

Remote load from a hot key with L1 caches and replication

$$R_{\text{remote}} \leq \min\left(\lambda_k, \frac{A}{T_{L1}}\right), \qquad R_{\text{per node}} = \frac{R_{\text{remote}}}{r}$$

Symbol	Meaning	Units
λ_k	Request rate for the hot key	requests/s
A	Number of app instances with an L1 cache	instances
T_{L1}	L1 TTL	s
r	Number of replicated copies of the key	copies
R_{remote}, $R_{\text{per node}}$	Requests to the remote tier, and to each node holding a copy	requests/s

Chapter 21 — Content Delivery Networks

From §21.3.1

FORMULA

Time to fetch an object over a new connection

$$T_{\text{fetch}} \approx (n_{\text{hs}} + \lceil \log_2 (\frac{S}{W_0} + 1) \rceil) \times \text{RTT} + t_{\text{server}}$$

Symbol	Meaning	Units
n_{hs}	Handshake round trips before the request (TCP + TLS 1.3 = 2; QUIC = 1; reused connection = 0)	round trips
S	Response size	bytes
W_0	Initial congestion window (typically 10 segments × 1,460 B ≈ 14.6 KB)	bytes
$\lceil \log_2 (S/W_0 + 1) \rceil$	Slow-start rounds to deliver S, the first of which carries the request (§7.7.7)	round trips
RTT	Round-trip time between client and server	s
t_{server}	Server processing time	s

From §21.5.1

FORMULA

Request and byte offload

$$O_R = 1 - \frac{N_{\text{origin}}}{N_{\text{total}}} \qquad O_B = 1 - \frac{V_{\text{origin}}}{V_{\text{total}}}$$

Symbol	Meaning	Units
O_R, O_B	Request offload, byte offload	dimensionless (0–1)
N_{origin}	Requests the CDN sent to the origin	requests
N_{total}	Requests users sent to the CDN	requests
V_{origin}	Bytes the origin sent to the CDN	bytes
V_{total}	Bytes the CDN sent to users	bytes

From §21.5.2

FORMULA

Origin fraction and origin bandwidth across tiers

$$f_{\text{origin}} = \prod_{i=1}^{k} (1 - h_i) \qquad B_{\text{origin}} = B_{\text{users}} \times (1 - O_B)$$

Symbol	Meaning	Units
h_i	Hit ratio of tier i, measured on the requests that *reach* that tier	dimensionless
k	Number of cache tiers (browser, edge, mid-tier, ...)	count
f_{origin}	Fraction of all requests that reach the origin, $1 - O_R$	dimensionless
B_{users}	Bandwidth delivered to users	Gb/s
B_{origin}	Bandwidth the origin must serve	Gb/s

From §21.5.3

FORMULA

Origin fetches per object per TTL window, without and with a shield

$$F_{\text{no shield}} \approx N(1 - e^{-\lambda T/N}) \qquad F_{\text{shield}} \approx 1$$

Symbol	Meaning	Units
N	Number of edge PoPs, with requests spread evenly among them	count
λ	Global request rate for the object	requests/s
T	Freshness lifetime (TTL) of the object	s
$1 - e^{-\lambda T/N}$	Probability that a given PoP sees at least one request in T (Poisson arrivals)	dimensionless
F	Expected origin fetches for the object per window T	fetches

From §21.5.4

FORMULA

Monthly delivery cost

$$C = V \times p_{\text{cdn}} + V \times (1 - O_B) \times p_{\text{origin}}$$

Symbol	Meaning	Units
V	Bytes delivered to users per month	GB
p_{cdn}	CDN delivery price	$/GB
p_{origin}	Origin egress price (cloud Internet egress)	$/GB
O_B	Byte offload	dimensionless
C	Monthly delivery cost (excluding request fees and commitments)	$

From §21.6.1

FORMULA

Freshness test in a shared cache

$$\text{fresh} \iff a < L, \qquad L = \begin{cases} \texttt{s-maxage} & \text{if present} \\ \texttt{max-age} & \text{otherwise} \end{cases}$$

Symbol	Meaning	Units
a	Current age of the stored response, including time spent in upstream caches (`Age`)	s
L	Freshness lifetime for a shared cache (RFC 9111 §4.2.1; `Expires` or heuristics apply if neither directive is present)	s

From §21.7.2

FORMULA

Time to first byte, direct versus through an edge

$$T_{\text{direct}} = (n_{\text{hs}} + 1)\,\text{RTT}_{uo} + t_{\text{app}} \qquad T_{\text{edge}} = (n_{\text{hs}} + 1)\,\text{RTT}_{ue} + \text{RTT}_{eo} + t_{\text{app}}$$

Symbol	Meaning	Units
n_{hs}	Client handshake round trips (2 for TCP + TLS 1.3, 0 if reused)	round trips
RTT_{uo}	Round trip, user to origin	ms
RTT_{ue}	Round trip, user to edge	ms
RTT_{eo}	Round trip, edge to origin over a warm, already-open connection	ms
t_{app}	Origin processing time	ms

From §21.8.2

FORMULA

Edge egress and storage for a bitrate ladder

$$B_{\text{edge}} = V \times \sum_r s_r b_r \qquad B_{\text{origin,live}} \approx M \times \sum_r b_r \qquad S_{\text{title}} = \frac{D}{8} \sum_r b_r$$

Symbol	Meaning	Units
V	Concurrent viewers	viewers
b_r	Bitrate of rendition r	Mb/s
s_r	Share of viewers on rendition r ($\sum s_r = 1$)	dimensionless
M	Number of shields (or PoPs, if unshielded) fetching each live rendition	count
D	Title duration	s
S_{title}	Stored size of one encoding of the full ladder	Mb ÷ 8 = MB

From §21.8.3

FORMULA

Throughput rule and buffer-based rate map

$$\hat{C} = \frac{n}{\sum_{i=1}^{n} 1/c_i}, \quad R_{\text{tput}} = \max\{b_r \le \beta\hat{C}\} \qquad R_{\text{BBA}}(x) = \begin{cases} b_{\min} & x \le x_r \\ b_{\min} + \frac{x - x_r}{x_c}(b_{\max} - b_{\min}) & x_r < x < x_r + x_c \\ b_{\max} & x \ge x_r + x_c \end{cases}$$

Symbol	Meaning	Units
c_i	Measured throughput of the i-th most recent segment download	Mb/s
n	Number of recent segments averaged	count
$\hat{C}$	Harmonic-mean throughput estimate	Mb/s
β	Safety factor (e.g. 0.8)	dimensionless
x	Current buffer level	s
x_r, x_c	Reservoir and cushion	s
$b_{\min}$, $b_{\max}$	Lowest and highest ladder rates; the BBA result is rounded down to a ladder rung	Mb/s

From §21.8.4

FORMULA

Glass-to-glass latency of segmented live video

$$L \approx t_{\text{enc}} + k \times D_{\text{seg}} + t_{\text{poll}}$$

Symbol	Meaning	Units
t_{enc}	Capture, encode and package delay	s
k	Segments (or parts) held back by the player	count
D_{seg}	Segment (or part) duration	s
t_{poll}	Delay before the player learns a new segment exists, plus download	s
L	Delay from camera to screen	s

Chapter 22 — API and Traffic Management

From §22.4.1

FORMULA

Token bucket state and admission bound

$$\text{tokens}(t) = \min\left(b,\ \text{tokens}(t_0) + r\,(t - t_0)\right) \qquad A(T) \le b + rT$$

Symbol	Meaning	Units
r	Refill rate (sustained allowed rate)	tokens/s (= requests/s at unit cost)
b	Bucket capacity (maximum burst)	tokens
t_0, t	Time of last update, current time	s
$A(T)$	Maximum requests admitted in any interval of length T	requests

From §22.5.2

FORMULA

GCRA (virtual scheduling form)

$$T = \frac{1}{r}, \qquad \tau = (b - 1)\,T, \qquad \text{admit at time } t \iff t \ge \text{TAT} - \tau, \qquad \text{then TAT} \leftarrow \max\left(\text{TAT}, t\right) + T$$

Symbol	Meaning	Units
r	Sustained allowed rate	requests/s
T	Emission interval: ideal spacing between requests	s
b	Burst size (requests that may arrive back to back)	requests
τ	Tolerance: how early a request may be relative to its ideal schedule	s
TAT	Theoretical arrival time of the next request (the only stored state)	s (timestamp)
t	Arrival time of the current request	s (timestamp)

From §22.6.1

FORMULA

Worst-case burst of a fixed window

$$A_{\max}(W) = 2L \qquad \text{peak short-term rate} \approx \frac{2L}{\delta}$$

Symbol	Meaning	Units
L	Limit per window	requests
W	Window length	s
$A_{\max}(W)$	Most requests admitted in some interval of length W that straddles a boundary	requests
δ	Duration of the straddling burst (can be arbitrarily small)	s

From §22.7.2

FORMULA

Sliding window counter estimate

$$\hat{N}(t) = C_{\text{prev}} \cdot \frac{W - e}{W} + C_{\text{curr}}, \qquad \text{admit if } \hat{N}(t) + 1 \le L$$

Symbol	Meaning	Units
$\hat{N}(t)$	Estimated requests in the sliding interval $(t - W, t]$	requests
C_{prev}	Count in the previous fixed window	requests
C_{curr}	Count so far in the current fixed window	requests
e	Time elapsed since the current window started	s
W	Window length	s
L	Limit per sliding window	requests

From §22.8.3

FORMULA

Overshoot of local counting with synchronization interval Δ

$$E_{\max} \approx \lambda \cdot \Delta \qquad M_{\text{sync}} = \frac{N(N-1)}{\Delta} \text{ (all-to-all)} \quad \text{or} \quad \frac{2N}{\Delta} \text{ (via a store or coordinator)}$$

Symbol	Meaning	Units
$E_{\max}$	Worst-case requests admitted beyond the limit per window	requests
λ	Offered rate for the key across all nodes	requests/s
Δ	Synchronization interval	s
N	Number of gateway nodes	nodes
M_{sync}	Coordination messages per second (each message can carry all keys)	messages/s

From §22.9.1

FORMULA

Time until goodput collapses without shedding

$$q(t) = (\lambda - \mu)\, t, \qquad w(t) = \frac{q(t)}{\mu}, \qquad t_{\text{collapse}} = \frac{\mu\, T_{\text{out}}}{\lambda - \mu} \quad (\lambda > \mu)$$

Symbol	Meaning	Units
λ	Offered request rate	requests/s
μ	Server capacity (maximum completion rate)	requests/s
$q(t)$	Queue length after t seconds of overload (unbounded FIFO)	requests
$w(t)$	Queueing delay seen by a request arriving at t	s
T_{out}	Client timeout	s
t_{collapse}	Time after which every completed request is already too late	s

From §22.9.2

FORMULA

Client-side adaptive throttling (Google SRE)

$$P_{\text{reject}} = \max\left(0, \frac{\text{requests} - K \cdot \text{accepts}}{\text{requests} + 1}\right)$$

Symbol	Meaning	Units
requests	Requests the client attempted in the recent window	count
accepts	Requests the backend accepted (did not reject for overload)	count
K	Multiplier; how many requests per accept the client may still send (typically 2)	dimensionless
P_{reject}	Probability the client rejects a new request locally, without sending it	dimensionless

From §22.10

FORMULA

Bounding a queue by time (Little's Law)

$$Q_{\text{max}} = \mu \times D_{\text{queue}}$$

Symbol	Meaning	Units
Q_{max}	Maximum queue length	items
μ	Rate at which the consumer drains the queue	items/s
D_{queue}	Maximum acceptable time an item may wait in the queue	s

From §22.11

FORMULA

Timeout budget along a call chain

$$B_{i+1} = B_i - e_i - m_i, \qquad \text{start the call only if } B_{i+1} > \hat{s}_{i+1}$$

Symbol	Meaning	Units
B_i	Remaining budget when hop i receives the request	ms
e_i	Time hop i spends before calling hop $i+1$ (local work, earlier calls)	ms
m_i	Margin hop i reserves to build and return its own response	ms
B_{i+1}	Budget passed to hop $i+1$	ms
$\hat{s}_{i+1}$	Typical latency of hop $i+1$ (for example its p50)	ms

From §22.12.2

FORMULA

Backoff with jitter (attempt $k = 1, 2, \ldots$)

$$s_k = \min(\text{cap}, \text{base} \cdot 2^k)$$

$$\text{full: } U(0, s_k) \qquad \text{equal: } \frac{s_k}{2} + U(0, \frac{s_k}{2}) \qquad \text{decorrelated: } \min(\text{cap}, U(\text{base}, 3w_{k-1}))$$

Symbol	Meaning	Units
k	Retry number	count
base	Initial backoff unit	ms
cap	Maximum wait	ms
s_k	Exponential backoff ceiling for retry k	ms
$U(a, b)$	A uniformly random value between a and b	ms
w_{k-1}	The previous wait for decorrelated jitter ($w_0 = \text{base}$)	ms

From §22.12.3

FORMULA

Retry amplification

$$A = \prod_{i=1}^{d} a_i = a^d \text{ (identical layers)}, \qquad \mathbb{E}[\text{attempts per layer}] = \sum_{j=0}^{a-1} p^j = \frac{1-p^a}{1-p}$$

Symbol	Meaning	Units
A	Worst-case calls reaching the bottom layer per user request	calls
a_i	Maximum attempts (1 + retries) at layer i	attempts
d	Number of layers that retry	layers
p	Probability that one attempt fails ($p < 1$)	dimensionless

From §22.15.4

FORMULA

Idempotency-key storage

$$S = \lambda_w \times T_{\text{ttl}} \times s_r$$

Symbol	Meaning	Units
S	Steady-state storage for key records	bytes
λ_w	Rate of keyed (non-idempotent) requests	requests/s
T_{ttl}	Retention window	s
s_r	Average record size (key, fingerprint, stored response)	bytes

Chapter 23 — Performance Engineering

From §23.4.1

FORMULA

Sampling error of a profile

$$n = f \times T \times c \qquad \hat{p} = \frac{k}{n} \qquad \mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}}$$

Symbol	Meaning	Units
f	Sampling frequency per CPU	samples/second
T	Profiling duration	seconds
c	Number of busy CPUs sampled	count
n	Total samples collected	samples
k	Samples whose stack contains the function	samples
$\hat{p}$	Estimated fraction of CPU time in the function	dimensionless
SE	Standard error of the estimate (binomial approximation)	dimensionless

From §23.5.2

FORMULA

Interactive response time law (closed systems)

$$X = \frac{N}{R + Z} \qquad \Leftrightarrow \qquad R = \frac{N}{X} - Z$$

Symbol	Meaning	Units
N	Number of users (or load-generator threads) in the closed loop	count
X	Throughput	requests/second
R	Mean response time	seconds
Z	Mean think time between a response and the next request	seconds

From §23.5.3

FORMULA

Damage done by a stall in an open system

$$\text{missed by closed tester} \approx \frac{T_s}{\Delta} \qquad t_{\text{drain}} = \frac{\lambda T_s}{\mu - \lambda} \qquad \text{affected requests} = \lambda (T_s + t_{\text{drain}}) = \frac{\lambda \mu T_s}{\mu - \lambda}$$

Symbol	Meaning	Units
T_s	Duration of the stall	seconds
$\Delta = 1/\lambda$	Intended interval between requests	seconds
λ	Intended (open) arrival rate	requests/second
μ	Service rate once the server resumes ($> \lambda$)	requests/second
t_{drain}	Time to drain the backlog after the stall	seconds

From §23.5.5

FORMULA

HdrHistogram footprint (simplified from the reference implementation)

$$s = 2^{\lceil \log_2 (2 \times 10^d) \rceil} \qquad b = 1 + \lceil \log_2 \frac{H}{s \cdot L} \rceil \qquad \text{counters} = (b + 1)\frac{s}{2} \qquad \text{memory} \approx 8 \times \text{counters}$$

Symbol	Meaning	Units
d	Significant decimal digits preserved	digits
s	Sub-buckets per bucket	count
L, H	Lowest discernible and highest trackable values	e.g. µs
b	Number of power-of-two buckets	count
memory	Size with 64-bit counters	bytes

From §23.6.1

FORMULA

Percentiles of a fan-out request

$$P(T_{\text{root}} \le t) = F(t)^N \qquad t_q^{\text{root}} = F^{-1}(q^{1/N}) \qquad \text{exponential leaves: } t_q^{\text{root}} = m \ln \frac{1}{1 - q^{1/N}}$$

Symbol	Meaning	Units
$F(t)$	CDF of one leaf's latency	dimensionless
N	Number of leaves the root waits for	count
q	Desired root percentile (as a fraction)	dimensionless
t_q^{root}	Root latency at percentile q	ms
m	Mean leaf latency for exponentially distributed leaves	ms

From §23.6.3

FORMULA

Throttling under a CPU quota

$$t_{\text{run}} = \frac{Q}{k} \qquad t_{\text{throttled}} = P - \frac{Q}{k} \qquad (\text{when } k > Q/P)$$

Symbol	Meaning	Units
P	Enforcement period	ms
Q	CPU-time quota per period (2 CPUs = $2P$)	CPU·ms
k	Threads runnable at the same time	count
t_{run}	Wall time until the quota is exhausted	ms
$t_{\text{throttled}}$	Wall time the whole cgroup is frozen	ms

From §23.7.1

FORMULA

Hedged latency and extra load (independent replicas)

$$T = \min(X_1,\ d + X_2) \qquad P(T > t) = \begin{cases} P(X > t) & t < d \\ P(X > t)\,P(X > t - d) & t \geq d \end{cases} \qquad \text{extra load} = P(X > d)$$

Symbol	Meaning	Units
X_1, X_2	Latencies of the primary and the hedge, same distribution as X	ms
d	Hedge delay	ms
T	Latency seen by the client	ms
extra load	Fraction of requests that send a second copy	dimensionless

From §23.8.1

FORMULA

Batch size and latency with a time limit

$$B_{\text{eff}} \approx \min(B_{\max},\ \max(1, \lambda T_{\max})) \qquad X_{\max} = \frac{B_{\text{eff}}}{t_0 + B_{\text{eff}}\,t_1} \qquad W_{\text{worst}} \approx T_{\max} + t_0 + B_{\text{eff}}\,t_1$$

Symbol	Meaning	Units
λ	Item arrival rate	items/second
$B_{\max}, T_{\max}$	Size and time limits of the batcher	items, seconds
B_{eff}	Typical batch size actually sent	items
t_0, t_1	Fixed cost per batch, marginal cost per item	seconds, seconds/item
$X_{\max}$	Throughput one sender can sustain	items/second
W_{worst}	Worst-case latency added by batching and sending	seconds

From §23.8.2

FORMULA

Sequential versus pipelined requests

$$T_{\text{seq}} = n(r + s) \qquad T_{\text{pipe}} \approx r + ns \quad (\text{window} \geq n)$$

Symbol	Meaning	Units
n	Number of requests	count
r	Network round-trip time	seconds
s	Server processing time per request	seconds

From §23.8.3

FORMULA

When compression reduces transfer time (store-and-forward)

$$t_{\text{plain}} = \frac{S}{W} \qquad t_{\text{comp}} = \frac{S}{C_c} + \frac{S}{\rho W} + \frac{S}{C_d} \qquad \text{compress if } \frac{1}{C_c} + \frac{1}{\rho W} + \frac{1}{C_d} < \frac{1}{W}$$

Symbol	Meaning	Units
S	Uncompressed size	MB
W	Network (or disk) bandwidth	MB/s
ρ	Compression ratio (original / compressed)	dimensionless
C_c, C_d	Compression and decompression speed	MB/s

From §23.9.1

FORMULA

GC frequency and pause exposure

$$I_{\text{minor}} = \frac{E}{a} \qquad f_{\text{hit}} \approx \frac{p + w}{I_{\text{minor}}} \qquad \text{Go heap target} = H_{\text{live}} \times \left(1 + \frac{\text{GOGC}}{100}\right)$$

Symbol	Meaning	Units
E	Young-generation (Eden) size	MB
a	Allocation rate	MB/s
I_{minor}	Interval between minor collections	seconds
p	STW pause per minor collection	seconds
w	Request duration	seconds
f_{hit}	Approximate fraction of requests that overlap a pause	dimensionless
H_{live}	Live heap after the previous Go collection	MB
GOGC	Go's GC percentage setting (default 100)	percent

From §23.10.1

FORMULA

USL for measured throughput, and its linearized form

$$X(N) = \frac{\gamma N}{1 + \alpha(N - 1) + \beta N(N - 1)} \qquad \frac{\gamma N}{X(N)} - 1 = \alpha(N - 1) + \beta N(N - 1)$$

Symbol	Meaning	Units
N	Concurrency (client threads, cores or nodes)	count
$X(N)$	Measured throughput at concurrency N	requests/second
γ	Throughput at $N = 1$ (the slope of ideal linear scaling)	requests/second
α, β	Contention and coherency coefficients (§4.13)	dimensionless

From §23.10.3

FORMULA

Fleet size from per-instance capacity

$$n = \lceil \frac{\lambda_{\text{peak}} \, (1 + g)}{u \cdot X_{\text{max}}} \rceil \qquad n_{\text{zones}} = Z \cdot \lceil \frac{n}{Z - 1} \rceil$$

Symbol	Meaning	Units
λ_{peak}	Forecast peak request rate	requests/second
g	Growth allowance over the planning horizon	dimensionless
X_{max}	Fitted peak throughput of one instance	requests/second
u	Target fraction of X_{max} at peak (headroom for queueing and variance)	dimensionless
Z	Number of availability zones, sized to survive the loss of one	count

Chapter 24 — Distributed Systems Fundamentals

From §24.4.1

FORMULA

Availability of k independent replicas, and the chance that something in a fleet fails

$$A_k = 1 - (1 - a)^k \qquad P_{\text{any}} = 1 - (1 - p)^n$$

Symbol	Meaning	Units
a	Availability of one replica (fraction of time it can serve)	dimensionless, 0–1
k	Number of replicas, any one of which can serve a request	count
A_k	Availability of the replicated service, assuming independent failures and instant failover	dimensionless
p	Probability that one node fails during a given period	dimensionless
n	Number of nodes in the fleet	count
P_{any}	Probability that at least one node fails in that period	dimensionless

From §24.6.3

FORMULA

Minimum replicas for consensus in the partially synchronous model

$$n_{\text{crash}} \geq 2f + 1 \qquad n_{\text{Byz}} \geq 3f + 1$$

Symbol	Meaning	Units
f	Maximum number of nodes that may be faulty at once	count
n_{crash}	Nodes needed when faults are crashes or omissions	count
n_{Byz}	Nodes needed when faults may be arbitrary	count

From §24.8.3

FORMULA

Timeout from bounded delays, and its practical percentile form

$$T = 2d + r \qquad T_{\text{practical}} = \text{RTT}_{p99.9} + r_{p99.9} + m$$

Symbol	Meaning	Units
d	Upper bound on one-way network delay	s
r	Upper bound on the time the remote node takes to process the request	s
T	Timeout after which the absence of a reply proves a fault (only if the bounds are real)	s
$\text{RTT}_{p99.9}$	Measured 99.9th-percentile round-trip time	s
$r_{p99.9}$	Measured 99.9th-percentile processing time	s
m	Safety margin for pauses (garbage collection, scheduling)	s

From §24.9.2

FORMULA

Clock drift and the resynchronization interval

$$\Delta(t) = \rho\, t \qquad \Delta_{AB}(t) \le 2\rho\, t \qquad T_{\text{sync}} \le \frac{\varepsilon}{2\rho}$$

Symbol	Meaning	Units
ρ	Maximum drift rate of one clock	ppm (1 ppm = 10^{-6} s/s)
t	Time since the clock was last synchronized	s
$\Delta(t)$	Worst-case error of one clock relative to true time	s
$\Delta_{AB}(t)$	Worst-case skew between two clocks drifting in opposite directions	s
ε	Largest skew you can tolerate between any two clocks	s
T_{sync}	Longest interval between synchronizations that keeps skew within ε	s

From §24.9.3

FORMULA

NTP offset, round-trip delay and error bound (RFC 5905)

$$\theta = \frac{(T_2 - T_1) + (T_3 - T_4)}{2} \qquad \delta = (T_4 - T_1) - (T_3 - T_2) \qquad |\theta - \theta_{\text{true}}| \le \frac{\delta}{2}$$

Symbol	Meaning	Units
T_1	Client clock when the request is sent	s
T_2	Server clock when the request arrives	s
T_3	Server clock when the reply is sent	s
T_4	Client clock when the reply arrives	s
θ	Estimated offset: server clock minus client clock; the client adds θ to correct itself	s
δ	Round-trip network delay, excluding the server's processing time	s
θ_{true}	Actual offset	s

From §24.9.5

FORMULA

Leap-smear rate

$$r_{\text{smear}} = \frac{1\ \text{s}}{W} \qquad e_{\text{max}} = 0.5\ \text{s (midpoint offset from UTC)}$$

Symbol	Meaning	Units
W	Smear window	s
r_{smear}	Rate by which smeared clocks run slow during the window	ppm
e_{max}	Offset between a smeared clock and true UTC at the leap second instant	s

From §24.11.4

FORMULA

Vector clock metadata size

$$S = n(b_{\text{id}} + b_{\text{ctr}}) \qquad M = K \cdot S$$

Symbol	Meaning	Units
n	Number of entries (processes, replicas or clients that have written)	count
b_{id}	Bytes to identify an entry's owner	bytes
b_{ctr}	Bytes per counter	bytes
S	Metadata per version	bytes
K	Number of keys stored	count
M	Total version metadata	bytes

From §24.12.4

FORMULA

Read uncertainty interval

$$t_r < t_w \le t_r + \varepsilon_{\max} \;\Rightarrow\; \text{restart the read at } t_w$$

Symbol	Meaning	Units
t_r	The reading transaction's HLC timestamp	s
t_w	Commit timestamp of a value found by the read	s
$\varepsilon_{\max}$	Configured maximum clock offset between any two nodes	s

From §24.14.1

FORMULA

Undetected-corruption probability over many transfers

$$P_{\text{bad}} = 1 - (1 - q)^{N}$$

Symbol	Meaning	Units
q	Probability that one transfer suffers corruption no lower layer catches	dimensionless
N	Number of transfers	count
P_{bad}	Probability that at least one transfer is silently corrupted	dimensionless

Chapter 25 — Consistency and Availability

From §25.3.4

FORMULA

Latency of a quorum operation

$$T_{\text{op}} \approx k \times \text{RTT}_{(q)} + T_{\text{proc}}, \qquad q = \lfloor N/2 \rfloor + 1$$

Symbol	Meaning	Units
T_{op}	Latency of one linearizable read or write as seen by the client	ms
k	Sequential round trips the protocol needs (ABD read: 2; ABD single-writer write: 1; Raft write via leader: 1 to the followers plus the client-to-leader trip)	count
$\text{RTT}_{(q)}$	Round-trip time to the q-th closest replica, counting the coordinator itself if it holds a replica	ms
q	Majority quorum size for N replicas	replicas
N	Number of replicas	replicas
T_{proc}	Processing and disk flush time	ms

From §25.3.4

FORMULA

Lower bound for sequentially consistent and linearizable registers

$$t_{\text{read}} + t_{\text{write}} \geq d$$

Symbol	Meaning	Units
t_{read}	Worst-case time for a read to return	ms
t_{write}	Worst-case time for a write to return	ms
d	Maximum one-way message delay between replicas	ms

From §25.7.2

FORMULA

Probability that a partial-quorum read misses the latest write

$$p_s = \frac{\binom{N-W}{R}}{\binom{N}{R}}, \qquad P(\text{older than all } k \text{ most recent versions}) = p_s^k$$

Symbol	Meaning	Units
p_s	Probability that all R replicas read hold none of the latest write's W copies	probability
N	Replication factor	replicas
W	Replicas that acknowledged the write when it committed	replicas
R	Replicas whose responses the read waits for	replicas
k	Number of most recent versions, any one of which counts as fresh enough (the read may miss $k-1$ of them)	versions

From §25.9.2

FORMULA

DynamoDB read capacity for a read rate

$$\mathrm{RCU} = r \times \left\lceil \frac{s}{4\ \mathrm{KB}} \right\rceil \times f, \qquad f = \begin{cases} 1 & \text{strongly consistent} \\ 0.5 & \text{eventually consistent} \end{cases}$$

Symbol	Meaning	Units
RCU	Provisioned read capacity needed	RCU
r	Read rate	reads/s
s	Item size	KB
f	Consistency factor	RCU per 4 KB read

Chapter 26 — Consensus and Coordination

From §26.4.1

FORMULA

Majority quorum size and fault tolerance

$$q = \left\lfloor \frac{N}{2} \right\rfloor + 1 \qquad f = N - q = \left\lfloor \frac{N-1}{2} \right\rfloor \qquad \Leftrightarrow \qquad N = 2f + 1$$

Symbol	Meaning	Units
N	Number of voting nodes	nodes
q	Majority quorum size	nodes
f	Crash failures tolerated while still able to form a quorum	nodes

From §26.4.1

FORMULA

Availability of a majority-quorum cluster (independent failures)

$$A_{\text{cluster}} = \sum_{k=q}^{N} \binom{N}{k} a^{k} (1-a)^{N-k}$$

Symbol	Meaning	Units
A_{cluster}	Probability that a quorum of nodes is up	dimensionless
a	Availability of one node	dimensionless
N, q	Cluster size and quorum size	nodes
$\binom{N}{k}$	Number of ways to choose which k nodes are up	count

From §26.4.2

FORMULA

Quorum commit latency

$$L_{\text{commit}} \approx t_{\text{fsync}} + \text{RTT}_{(q-1)}$$

Symbol	Meaning	Units
L_{commit}	Time from the leader receiving a write to the entry being committed	ms
t_{fsync}	Time to flush the entry to durable storage (leader and followers flush in parallel)	ms
$\text{RTT}_{(k)}$	The k-th smallest round-trip time from the leader to its followers, including the followers' flush	ms
q	Quorum size	nodes

From §26.4.3

FORMULA

Flexible Paxos quorum condition

$$|Q_1| + |Q_2| > N$$

Symbol	Meaning	Units
$\lvert Q_1 \rvert$	Size of a leader-election (phase-1) quorum	nodes
$\lvert Q_2 \rvert$	Size of a replication (phase-2) quorum	nodes
N	Number of acceptors	nodes

From §26.7.2

FORMULA

Raft election timing and expected failover time

$$h \approx \mathrm{RTT}_{\mathrm{typ}}, \qquad T_{\min} \geq 10 \times \mathrm{RTT}_{\mathrm{typ}}, \qquad T_{\mathrm{failover}} \approx T_{\min} + \frac{W}{n+1} - \frac{h}{2} + \mathrm{RTT} + t_{\mathrm{fsync}}$$

Symbol	Meaning	Units
h	Heartbeat interval	ms
$\mathrm{RTT}_{\mathrm{typ}}$	Typical round-trip time between members	ms
$T_{\min}, W$	Election timeout is drawn uniformly from $[T_{\min}, T_{\min} + W]$	ms
n	Followers still alive after the leader fails	nodes
T_{failover}	Expected time from leader crash to a new leader	ms

From §26.7.2

FORMULA

Probability that a second candidate starts before the first's votes arrive

$$P_{\text{contend}} = 1 - (1 - \frac{\delta}{W})^{n}$$

Symbol	Meaning	Units
δ	Time for the first candidate's RequestVote to reach the others (one-way delay plus flush)	ms
W	Width of the randomized timeout range	ms
n	Number of followers whose timers are running	nodes
P_{contend}	Probability that the gap between the first two timeouts is below δ	dimensionless

From §26.11.1

FORMULA

Lease safety margin

$$T_{\text{holder}} = T(1 - \rho) - d_{\text{max}} \qquad T_{\text{grantor}} = T(1 + \rho)$$

Symbol	Meaning	Units
T	Nominal lease duration	s
ρ	Maximum clock rate error relative to real time (drift bound)	dimensionless (e.g. 10^{-4} = 100 ppm)
d_{max}	Maximum time between the holder checking validity and its action taking effect (processing, network, budgeted pause)	s
T_{holder}	Local elapsed time after which the holder must stop acting	s
T_{grantor}	Local elapsed time after grant before the grantor may grant the right to someone else	s

Chapter 27 — Replication and Fault Tolerance

From §27.4.1

FORMULA

Writes lost in an asynchronous failover

$$L = r_w \times \ell$$

Symbol	Meaning	Units
L	Acknowledged writes lost when the most up-to-date backup is promoted	writes
r_w	Write rate on the primary just before the crash	writes/s
ℓ	Replication lag of the promoted backup at the moment of the crash	s

From §27.5.2

FORMULA

Chain write latency and read capacity

$$T_{\text{write}} \approx (n+1)\,d + n\,s \qquad X_{\text{read}}^{\text{CR}} = c \qquad X_{\text{read}}^{\text{CRAQ}} \approx n\,c \text{ (clean reads)}$$

Symbol	Meaning	Units
n	Number of nodes in the chain	nodes
d	One-way network delay per hop	ms
s	Processing time per node for an update	ms
c	Read capacity of one node	reads/s
T_{write}	Client-observed write latency (client → head → … → tail → client)	ms
X_{read}	Read capacity of one chain	reads/s

From §27.6.1

FORMULA

Conflict rate for one item written at two leaders

$$\kappa \approx 2\,\lambda_A\,\lambda_B\,\Delta \qquad (\lambda_A\Delta \ll 1,\ \lambda_B\Delta \ll 1)$$

Symbol	Meaning	Units
κ	Expected conflicts per second for the item	conflicts/s
λ_A, λ_B	Write rate to the item at leaders A and B	writes/s
Δ	Replication delay between the leaders	s

From §27.8.1

FORMULA

Quorum overlap and fault tolerance

$$|S_R \cap S_W| \ \geq\ R + W - N \qquad f_{\text{read}} = N - R \qquad f_{\text{write}} = N - W$$

Symbol	Meaning	Units
N	Replicas per key (replication factor)	replicas
R	Responses a read waits for	replicas
W	Acknowledgments a write waits for	replicas
S_R, S_W	The sets of replicas that answered a read and acknowledged a write	sets
$f_{\text{read}}, f_{\text{write}}$	Unavailable replicas a read or write can tolerate	replicas

From §27.8.3

FORMULA

Latency of the k-th fastest of N replicas (independent, identically distributed)

$$P(T_{(k)} > t) = \sum_{j=0}^{k-1} \binom{N}{j} F(t)^{j} \, (1 - F(t))^{N-j} \qquad \mathbb{E}[T_{(k)}] = \mu \sum_{i=N-k+1}^{N} \frac{1}{i} \text{ (exponential)}$$

Symbol	Meaning	Units
$T_{(k)}$	Time until the k-th reply arrives	ms
$F(t)$	Probability that one replica replies within t	dimensionless
N, k	Replicas contacted; replies required	count
j	Number of replicas that replied within t (summation index)	count
μ	Mean latency of one replica, if exponentially distributed	ms

From §27.9.1

FORMULA

Hint volume during an outage

$$V_h = \frac{r_w \cdot N}{n} \cdot T_{\text{down}} \cdot s$$

Symbol	Meaning	Units
V_h	Total hints accumulated for the failed node across the cluster	bytes
r_w	Cluster-wide client write rate	writes/s
N	Replication factor	replicas
n	Nodes in the cluster	nodes
T_{down}	Time the node is down (capped by the hint window)	s
s	Size of one write including overhead	bytes

From §27.10.2

FORMULA

Phi-accrual suspicion level (normal model)

$$\varphi(t_{\text{now}}) = -\log_{10}\left(1 - F(t_{\text{now}} - t_{\text{last}})\right), \qquad F(\delta) = \Phi\left(\frac{\delta - \mu}{\sigma}\right)$$

Symbol	Meaning	Units
φ	Suspicion level; the detector's false-suspicion probability is about $10^{-\varphi}$	dimensionless
$t_{\text{now}} - t_{\text{last}}$	Time since the last heartbeat arrived, δ	ms
μ, σ	Mean and standard deviation of recent inter-arrival times (sliding window)	ms
F	Estimated cumulative distribution of inter-arrival times	dimensionless
Φ	Standard normal cumulative distribution function	dimensionless

From §27.10.3

FORMULA

Rounds for push gossip to reach every node

$$T_{\text{rounds}} \approx \log_{1+f} n + \frac{\ln n}{f} \qquad (\text{for } f = 1:\ \log_2 n + \ln n + O(1))$$

Symbol	Meaning	Units
T_{rounds}	Rounds until every node is informed	rounds
n	Number of nodes	nodes
f	Peers contacted per node per round	peers/round

From §27.10.3

FORMULA

SWIM expected time to first detection

$$\mathbb{E}[T_{\text{detect}}] = \frac{T'}{1 - (1 - \frac{1}{n-1})^{n-1}} \quad \overset{n \to \infty}{\to} \quad T' \cdot \frac{e}{e-1} \approx 1.58\,T'$$

Symbol	Meaning	Units
T'	Protocol period	s
n	Group size	members
$\mathbb{E}[T_{\text{detect}}]$	Expected time until some member first probes the failed member	s

From §27.11.1

FORMULA

Failover time budget

$$T_{\text{fo}} = T_{\text{detect}} + T_{\text{elect}} + T_{\text{reconfig}} + T_{\text{redirect}} \qquad D_{\text{yr}} = v \cdot T_{\text{fo}}$$

Symbol	Meaning	Units
T_{fo}	Time from crash until clients write successfully again	s
T_{detect}	Time to suspect the failure	s
T_{elect}	Time to choose a replacement	s
T_{reconfig}	Time to fence, catch up and promote	s
T_{redirect}	Time until clients reach the replacement	s
v	Failovers per year	1/year
D_{yr}	Write downtime per year from failovers	s/year

From §27.12.1

FORMULA

Recovery time for snapshot plus log

$$T_{\text{rec}} = \frac{D}{B_{\text{eff}}} + \frac{r_{\text{in}} \cdot D/B_{\text{eff}}}{r_{\text{apply}} - r_{\text{in}}}, \qquad B_{\text{eff}} = \min\left(\sum_i B_{\text{src},i},\ B_{\text{dst}}\right)$$

Symbol	Meaning	Units
T_{rec}	Time until the replica is current	s
D	Size of the snapshot or data to copy	bytes
B_{eff}	Effective copy bandwidth	bytes/s
$B_{\text{src},i}$	Throttled send rate of source i	bytes/s
B_{dst}	Receiver limit: NIC or disk write rate	bytes/s
r_{in}	Rate of new log entries during the copy	entries/s
r_{apply}	Rate at which the replica replays entries	entries/s

From §27.14.2

FORMULA

Comparing version vectors

$$V \le V' \iff \forall i\colon V[i] \le V'[i] \qquad V \parallel V' \iff V \not\le V' \ \wedge\ V' \not\le V \qquad \text{merge: } (V \sqcup V')[i] = \max(V[i], V'[i])$$

Symbol	Meaning	Units
V, V'	Version vectors of two versions of the same item	counters per replica
$V[i]$	Number of writes to the item coordinated by replica i that this version has seen	count
$V \le V'$	V' descends from V, so V can be discarded	—
$V \parallel V'$	The versions are concurrent: keep both as siblings	—
$\sqcup$	Element-wise maximum, the vector of a merged version	—

Chapter 28 — Distributed Transactions

From §28.4.3

FORMULA

Exposure to in-doubt blocking

$$w \approx \mathrm{RTT} + t_{f,c} \qquad P_{\mathrm{doubt}} = 1 - e^{-\lambda w} \approx \lambda w \qquad N_{\mathrm{doubt}} = X w$$

Symbol	Meaning	Units
w	Window from the first YES vote reaching the coordinator's path until all participants hold the decision: half an RTT for votes, the coordinator's forced write, half an RTT for decisions	seconds
RTT	Network round-trip time between coordinator and participants	seconds
$t_{f,c}$	Time of the coordinator's forced log write	seconds
λ	Coordinator crash rate	crashes/second
P_{doubt}	Probability a given transaction is caught in doubt by a coordinator crash	dimensionless
X	Rate of distributed transactions through this coordinator	transactions/second
N_{doubt}	Expected number of transactions left in doubt by one crash (Little's Law, §4.11)	transactions

From §28.4.4

FORMULA

Two-phase commit latency and participant lock hold time

$$T_{\text{2PC}} \approx 2\,\text{RTT} + t_{f,p} + t_{f,c} \qquad T_{\text{hold}} \approx t_{f,p} + \text{RTT} + t_{f,c}$$

Symbol	Meaning	Units
T_{2PC}	Commit latency until every participant has received the decision	seconds
RTT	Round trip between coordinator and the slowest participant	seconds
$t_{f,p}$	Slowest participant's forced write of its PREPARED record (writes happen in parallel)	seconds
$t_{f,c}$	Coordinator's forced write of the COMMIT record	seconds
T_{hold}	Extra time a participant holds locks during commit, from forcing PREPARED until the decision arrives	seconds

From §28.4.4

FORMULA

Availability of a transaction spanning k partitions

$$A_{\text{txn}} = A_c \prod_{i=1}^{k} A_i \approx A^k \quad \text{(equal partitions, coordinator co-located with one of them)}$$

Symbol	Meaning	Units
A_{txn}	Probability that the transaction can commit at a random moment	dimensionless
A_c	Availability of the coordinator and its log	dimensionless
A_i	Availability of partition i	dimensionless
k	Number of partitions the transaction touches	count

From §28.7.3

FORMULA

Percolator commit latency

$$T_{\text{perc}} \approx 2\,\text{RTT}_{\text{TSO}} + 2\,(\text{RTT} + t_w) \qquad T_{\text{async}} \approx \text{RTT}_{\text{TSO}} + \text{RTT} + t_w$$

Symbol	Meaning	Units
T_{perc}	Timestamp and commit overhead of a classic Percolator transaction (start and commit timestamps, prewrite, primary commit)	ms
T_{async}	The same with async commit (no commit timestamp fetch, no wait for the primary commit)	ms
RTT_{TSO}	Round trip to the timestamp oracle	ms
RTT	Round trip from the client to the storage node	ms
t_w	Durable, replicated write on the storage node (Raft quorum plus `fsync`)	ms

From §28.8.2

FORMULA

Commit wait

$$T_{\text{wait}} \approx 2\,\bar{\varepsilon} \qquad T_{\text{commit}} \approx \max\,(T_{\text{Paxos}},\ 2\,\bar{\varepsilon})$$

Symbol	Meaning	Units
T_{wait}	Time from choosing s = `TT.now().latest` until `TT.after(s)` holds	ms
$\bar{\varepsilon}$	Average TrueTime uncertainty (half-width) during the wait	ms
T_{Paxos}	Time to replicate the commit record to a quorum	ms
T_{commit}	Commit latency at the coordinator leader, ignoring 2PC phase 1	ms

From §28.9.1

FORMULA

Calvin transaction latency

$$T_{\text{Calvin}} \approx \frac{e}{2} + T_{\text{rep}} + T_{\text{exec}}$$

Symbol	Meaning	Units
e	Epoch length; a request waits on average half an epoch to be batched	ms
T_{rep}	Time to replicate the batch (Paxos round, or 0 for asynchronous)	ms
T_{exec}	Lock wait, read forwarding and execution	ms
T_{Calvin}	Expected time from request to result	ms

From §28.10.1

FORMULA

Commit latency in consensus rounds

$$T_{\text{serial}} \approx 2\,T_{\text{c}} \qquad T_{\text{parallel}} \approx T_{\text{c}}$$

Symbol	Meaning	Units
T_{c}	One consensus (Raft) round for the slowest range	ms
T_{serial}	Commit with intents then a committed record	ms
T_{parallel}	Commit with parallel commits	ms

Chapter 29 — Distributed Data Processing

From §29.3.1

FORMULA

Scan time of a bounded dataset

$$T_{\text{scan}} = \frac{D}{N \cdot b}$$

Symbol	Meaning	Units
T_{scan}	Time to read the whole input once, ignoring stragglers	s
D	Input size	bytes
N	Number of nodes reading in parallel	count
b	Sustained read-and-process rate per node	bytes/s

From §29.4.2

FORMULA

MapReduce job time (first-order estimate)

$$T_{\text{job}} \approx T_0 + \lceil \frac{M}{N s_m} \rceil \frac{C}{r_m} + \frac{\sigma D}{N B} + \lceil \frac{R}{N s_r} \rceil \frac{\sigma D / R}{r_r}, \qquad M = \lceil \frac{D}{C} \rceil$$

Symbol	Meaning	Units
T_0	Job start-up and commit overhead	s
D	Input size	bytes
C	Split size	bytes
M, R	Number of map and reduce tasks	count
N	Worker nodes	count
s_m, s_r	Map and reduce slots (concurrent tasks) per node	count
r_m, r_r	Per-task processing rate of map and reduce	bytes/s
σ	Shuffle ratio: map output bytes ÷ input bytes, after any combiner	dimensionless
B	Effective per-node shuffle network bandwidth	bytes/s

From §29.4.2

FORMULA

Shuffle volume with and without a combiner

$$V_{\text{no comb}} = n_{\text{pairs}} \cdot b_{\text{pair}}, \qquad V_{\text{comb}} \approx M \cdot u \cdot b_{\text{pair}}$$

Symbol	Meaning	Units
V	Bytes moved across the network in the shuffle	bytes
n_{pairs}	Total intermediate pairs emitted by all mappers	count
b_{pair}	Average serialized size of one key–value pair	bytes
M	Map tasks	count
u	Distinct keys per map task after combining ($u \leq$ pairs per task)	count

From §29.4.3

FORMULA

Probability that a phase contains at least one straggler

$$P(\text{at least one straggler}) = 1 - (1 - p)^{n} \approx 1 - e^{-np}$$

Symbol	Meaning	Units
p	Probability that any one task is a straggler	probability
n	Number of tasks in the phase (or in its last wave)	count

From §29.4.4

FORMULA

Network cost of a shuffle join versus a broadcast join

$$V_{\text{shuffle}} \approx (|L| + |S|) \cdot \frac{N-1}{N}, \qquad V_{\text{broadcast}} \approx |S| \cdot (N-1)$$

Broadcast is cheaper when $|S| \cdot N < |L| + |S|$, roughly $|S| < |L|/(N-1)$, and $|S|$ fits in each executor's memory.

Symbol	Meaning	Units
$\lvert L\rvert, \lvert S\rvert$	Size of the large and small input	bytes
N	Number of executors or nodes receiving data	count
V	Bytes sent over the network	bytes

From §29.5.3

FORMULA

Speedup from caching an iterative job

$$S = \frac{k\,(T_{\text{load}} + T_{\text{iter}})}{T_{\text{load}} + k\,T_{\text{iter}}} \;\xrightarrow{k \to \infty}\; 1 + \frac{T_{\text{load}}}{T_{\text{iter}}}$$

Symbol	Meaning	Units
S	Speedup of the cached version over reloading every iteration	dimensionless
k	Number of iterations (or repeated queries)	count
T_{load}	Time to read, parse and build the working dataset from storage	s
T_{iter}	Compute time of one iteration over the in-memory data	s

From §29.6.2

FORMULA

Shuffle time lower bound on a rack-structured cluster

$$T_{\text{shuffle}} \approx \max\left(\frac{V(N-1)/N}{N\eta B_{\text{NIC}}},\ \frac{V(K-1)/K}{KU},\ \frac{2V}{NB_{\text{disk}}}\right)$$

Symbol	Meaning	Units
V	Shuffled data volume	bytes
N, K	Nodes and racks	count
B_{NIC}	Network interface speed per node	bytes/s
η	Achievable fraction of line rate	dimensionless
U	Uplink capacity per rack	bytes/s
B_{disk}	Local disk throughput per node (each byte written once, read once)	bytes/s

From §29.6.3

FORMULA

Slowdown from a single hot key

$$\text{skew} = \frac{\max_i L_i}{\bar{L}} \approx fP + (1-f), \qquad \text{with salt } s\text{: skew} \approx \frac{fP}{s} + (1-f)$$

Symbol	Meaning	Units
L_i, $\bar{L}$	Records in partition i; mean records per partition	records
f	Fraction of all records carrying the hot key	dimensionless
P	Number of shuffle partitions	count
s	Salt range (hot key spread over s partitions), $s \le P$	count

From §29.8.3

FORMULA

Fraction of events that miss their tumbling window

$$f_{\text{late}} \approx \frac{\mu}{W} e^{-\delta/\mu}\left(1 - e^{-W/\mu}\right) \qquad \text{added output latency} \approx \delta$$

Derivation: with exponentially distributed delay d (mean μ) and the watermark close to arrival time minus δ, an event at offset u before its window's end misses the window if $d > \delta + u$; averaging $e^{-(\delta+u)/\mu}$ over u uniform on $[0, W)$ gives the formula.

Symbol	Meaning	Units
f_{late}	Fraction of events arriving after their window fired	dimensionless
μ	Mean delay between event time and arrival	s
δ	Watermark delay (bounded out-of-orderness)	s
W	Window size	s

From §29.9.2

FORMULA

Checkpoint interval: overhead versus recovery

$$T_{\text{opt}} \approx \sqrt{2CM_f}, \qquad T_{\text{catch-up}} \approx \frac{\lambda(T/2 + T_{\text{restart}} + T_{\text{restore}})}{\mu - \lambda}$$

Symbol	Meaning	Units
T, T_{opt}	Checkpoint interval; interval minimizing total lost time (Young's approximation)	s
C	Throughput-equivalent cost of one checkpoint	s
M_f	Mean time between failures of the job	s
λ	Input rate	events/s
μ	Maximum processing rate of the job	events/s
T_{restart}, T_{restore}	Time to reschedule tasks; time to download state	s
$T_{\text{catch-up}}$	Time to work off the backlog after restoring	s